



COAL INDIA LIMITED

(A Govt. of India Undertaking)

(A Maharatna Company)

Recruitment of Management Trainees through Computer Based Test (CBT-26)

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Test Date	24/08/2026
Test Time	3:00 PM - 6:00 PM
Subject	System

Section : Part 1 General Knowledge Awareness

Q.1 During a National Emergency under Article 352, which Fundamental Right remains protected and cannot be suspended, as guaranteed by the 44th Amendment Act, 1978?

Ans ☒ A. Right to Freedom of Religion

☒ B. Right to Life and Personal Liberty

☒ C. Right to Freedom of Speech

☒ D. Right to Acquire Property

Question ID : 4410092339441

Option 1 ID : 4410099199863

Option 2 ID : 4410099199862

Option 3 ID : 4410099199860

Option 4 ID : 4410099199861

Status : Answered

Chosen Option : A

Q.2 Which former Indian tennis player received the Padma Bhushan in 2026 in the field of Sports?

Ans ☒ A. Vijay Amritraj

☐ B. Rohan Bopanna

☐ C. Mahesh Bhupathi

☐ D. Leander Paes

Question ID : 4410092216363

Option 1 ID : 4410098721586

Option 2 ID : 4410098721589

Option 3 ID : 4410098721588

Option 4 ID : 4410098721587

Status : Answered

Chosen Option : D

Q.3 Which of the following statements about the Tropical Rainforest Climate (Af) is correct?

Ans ☐ A. The region experiences temperatures below 10°C during winter months

☐ B. It experiences a distinct dry season lasting 3-4 months

☒ C. high temperatures and significant rainfall throughout the entire year

☐ D. It is found exclusively in the Amazon Basin

Question ID : 4410092216051

Option 1 ID : 4410098720341

Option 2 ID : 4410098720338

Option 3 ID : 4410098720339

Option 4 ID : 4410098720340

Status : Answered

Chosen Option : C

Q.4 Which of the following correctly describes the role of the 'Tropical Easterly Jet Stream' (TEJ) in the Indian monsoon system?

Ans ☒ A. It flows from east to west over peninsular India during summer, strengthening the southwest monsoon

☐ B. It flows from west to east over northern India during winter, bringing cold waves

☐ C. It flows from east to west over the Himalayas during winter, causing snowfall in Kashmir

☐ D. It flows from west to east over the southern peninsula during summer, inhibiting cloud formation

Question ID : 4410092216037

Option 1 ID : 4410098720283

Option 2 ID : 4410098720282

Option 3 ID : 4410098720285

Option 4 ID : 4410098720284

Status : Answered

Chosen Option : A

Q.5 The Revolt of 1857 broke out first at which military cantonment town?

Ans ☒ A. Kanpur

☒ B. Jhansi

☒ C. Lucknow

☒ D. Meerut

Question ID : 4410092215553

Option 1 ID : 4410098718353

Option 2 ID : 4410098718350

Option 3 ID : 4410098718351

Option 4 ID : 4410098718352

Status : Answered

Chosen Option : B

Q.6 The SHE-Marts scheme announced in Union Budget 2026–27 seeks to improve market access for products made by which group?

Ans ☒ A. Farmers under FPO networks

☒ B. Women from Self-Help Groups

☒ C. Students in rural schools

☒ D. Workers in urban factories

Question ID : 4410092216368

Option 1 ID : 4410098721608

Option 2 ID : 4410098721609

Option 3 ID : 4410098721607

Option 4 ID : 4410098721606

Status : Answered

Chosen Option : B

Q.7 The Pamban Bridge, India's first sea bridge, connects the mainland of Tamil Nadu to which of the following islands?

Ans ☒ A. Andaman Islands

☒ B. Rameswaram Island

☒ C. Majuli Island

☒ D. Diu Island

Question ID : 4410092216046

Option 1 ID : 4410098720320

Option 2 ID : 4410098720321

Option 3 ID : 4410098720318

Option 4 ID : 4410098720319

Status : Answered

Chosen Option : B

Q.8 Which regional organisation celebrated its 29th anniversary in June 2026, reaffirming cooperation in the Bay of Bengal region?

- Ans**
- ☒ A. ASEAN
- ☒ B. SCO
- ☒ C. SAARC
- ☒ D. BIMSTEC

Question ID : 4410092216358

Option 1 ID : 4410098721567

Option 2 ID : 4410098721566

Option 3 ID : 4410098721568

Option 4 ID : 4410098721569

Status : Answered

Chosen Option : D

Q.9 The 'October Heat' phenomenon in India is attributed to which of the following conditions?

- Ans**
- ☒ A. Low humidity and frequent cyclonic storms
- ☒ B. Intense low pressure over the Bay of Bengal
- ☒ C. Cold northerly winds from the Himalayas
- ☒ D. Clear skies, high insolation, and moist air from retreating monsoon

Question ID : 4410092216038

Option 1 ID : 4410098720286

Option 2 ID : 4410098720289

Option 3 ID : 4410098720288

Option 4 ID : 4410098720287

Status : Answered

Chosen Option : D

Q.10 Which scientist is credited with developing the smallpox vaccine, laying the foundation for immunology?

- Ans**
- ☒ A. Alexander Fleming
- ☒ B. Jonas Salk
- ☒ C. Paul Ehrlich
- ☒ D. Edward Jenner

Question ID : 4410092216056

Option 1 ID : 4410098720361

Option 2 ID : 4410098720358

Option 3 ID : 4410098720360

Option 4 ID : 4410098720359

Status : Answered

Chosen Option : D

Q.11 Which Parliamentary Committee is often called a 'mini-Parliament' as it scrutinises government spending using audit reports of the Comptroller and Auditor General?

- Ans**
- ☒ A. Public Undertakings Committee
- ☒ B. Public Accounts Committee
- ☒ C. Estimates Committee Report
- ☒ D. Business Advisory Committee

Question ID : 4410092339437

Option 1 ID : 4410099199846

Option 2 ID : 4410099199845

Option 3 ID : 4410099199844

Option 4 ID : 4410099199847

Status : Answered

Chosen Option : B

Q.12 What role does encryption play in application security?

- Ans**
- ☒ A. It protects information from unauthorized access
- ☒ B. It enhances display quality
- ☒ C. It increases processing power
- ☒ D. It slows down cyber attacks

Question ID : 4410091838568

Option 1 ID : 4410097247822

Option 2 ID : 4410097247824

Option 3 ID : 4410097247825

Option 4 ID : 4410097247823

Status : Answered

Chosen Option : A

Q.13 Which temple had been associated with the 41-day Mandala kaalam as its major pilgrimage season, followed by the Makaravilakku festival, and was surrounded by mountains and dense forests forming part of the Periyar Tiger Reserve?

- Ans**
- ☒ A. Sree Padmanabhaswamy Temple
- ☒ B. Thirunelli Temple
- ☒ C. Sabarimala Sree Dharma Sastha Temple
- ☒ D. Guruvayur Sree Krishna Temple

Question ID : 4410092215686

Option 1 ID : 4410098718883

Option 2 ID : 4410098718882

Option 3 ID : 4410098718884

Option 4 ID : 4410098718885

Status : Answered

Chosen Option : B

Q.14 What is the primary purpose of virtual memory in a computer system?

- Ans**
- ☒ A. To increase physical storage capacity
 - ☒ B. To extend RAM capacity by using disk space
 - ☒ C. To increase physical storage capacity
 - ☒ D. To improve graphics performance

Question ID : 441009296694

Option 1 ID : 4410091153880

Option 2 ID : 4410091153882

Option 3 ID : 4410091153881

Option 4 ID : 4410091153883

Status : Answered

Chosen Option : B

Q.15 Which of the following is the largest coal-based thermal power station in India by installed capacity as of March 2026?

- Ans**
- ☒ A. Vindhyachal STPS
 - ☒ B. Talcher STPS
 - ☒ C. Sipat Power Stn
 - ☒ D. Mundra Power Stn

Question ID : 4410092215542

Option 1 ID : 4410098718306

Option 2 ID : 4410098718308

Option 3 ID : 4410098718309

Option 4 ID : 4410098718307

Status : Answered

Chosen Option : A

Q.16 In which Union Budget had the Export Promotion Mission (EPM) been announced as a flagship structural reform aimed at strengthening India's trade competitiveness?

- Ans**
- ☒ A. Union Budget 2024–25
 - ☒ B. Union Budget 2023–24
 - ☒ C. Union Budget 2026–27
 - ☒ D. Union Budget 2025–26

Question ID : 4410092215680

Option 1 ID : 4410098718859

Option 2 ID : 4410098718858

Option 3 ID : 4410098718861

Option 4 ID : 4410098718860

Status : Answered

Chosen Option : A

Q.17 According to the IMF's World Economic Outlook released in April 2026, the global economy faced renewed disruption primarily due to which of the following developments?

- Ans**
- ☒ A. Breakdown of WTO rules
 - ☒ B. War in the Middle East
 - ☒ C. Collapse of global banking
 - ☒ D. Default by oil exporters

Question ID : 4410092216353

Option 1 ID : 4410098721548

Option 2 ID : 4410098721546

Option 3 ID : 4410098721547

Option 4 ID : 4410098721549

Status : Answered

Chosen Option : B

Q.18 In which year had the Goods and Services Tax (GST) been introduced in India, replacing a fragmented and complex system of indirect taxes?

- Ans**
- ☒ A. 2019
 - ☒ B. 2017
 - ☒ C. 2014
 - ☒ D. 2010

Question ID : 4410092215683

Option 1 ID : 4410098718872

Option 2 ID : 4410098718871

Option 3 ID : 4410098718870

Option 4 ID : 4410098718873

Status : Answered

Chosen Option : B

Q.19 Which Constitutional Amendment Act had enabled the creation of the Goods and Services Tax Council and conferred simultaneous power upon Parliament and the State Legislatures to make laws governing goods and services tax?

- Ans**
- ☒ A. 102nd Constitution Amendment Act, 2018
 - ☒ B. 103rd Constitution Amendment Act, 2019
 - ☒ C. 101st Constitution Amendment Act, 2016
 - ☒ D. 100th Constitution Amendment Act, 2015

Question ID : 4410092215674

Option 1 ID : 4410098718836

Option 2 ID : 4410098718837

Option 3 ID : 4410098718835

Option 4 ID : 4410098718834

Status : Answered

Chosen Option : C

Q.20 The 11th Governing Council Meeting held in June 2026 was associated with which Indian administrative and economic body?

Ans ☒ A. Election Commission

☒ B. NITI Aayog

☒ C. Central Vigilance Commission

☒ D. Finance Commission

Question ID : 4410092216361

Option 1 ID : 4410098721581

Option 2 ID : 4410098721579

Option 3 ID : 4410098721578

Option 4 ID : 4410098721580

Status : Answered

Chosen Option : B

Q.21 The PSLV-C62 / EOS-N1 Mission of ISRO, launched in January 2026, was conducted as the ninth dedicated commercial mission of which organisation?

Ans ☒ A. Indian National Space Promotion and Authorisation Centre

☒ B. Defence Research and Development Organisation

☒ C. NewSpace India Limited

☒ D. National Remote Sensing Centre

Question ID : 4410092216373

Option 1 ID : 4410098721628

Option 2 ID : 4410098721627

Option 3 ID : 4410098721626

Option 4 ID : 4410098721629

Status : Answered

Chosen Option : A

Q.22 Which ministry had released the SDG Bulletins titled "Planet in Focus: Advancing Environmental Sustainability under the SDGs" and "Delivering Prosperity at Scale: India's Economic Transformation through the SDGs" in March 2026?

Ans ☒ A. Ministry of Finance

☒ B. Ministry of Rural Development

☒ C. Ministry of Environment, Forest and Climate Change

☒ D. Ministry of Statistics and Programme Implementation

Question ID : 4410092215627

Option 1 ID : 4410098718648

Option 2 ID : 4410098718649

Option 3 ID : 4410098718647

Option 4 ID : 4410098718646

Status : Answered

Chosen Option : C

Q.23 Who founded the Mughal Empire in India after defeating Ibrahim Lodi at the First Battle of Panipat in 1526?

- Ans**
- ☒ A. Akbar
- ☒ B. Babur
- ☒ C. Humayun
- ☒ D. Jahangir

Question ID : 4410092215551

Option 1 ID : 4410098718343

Option 2 ID : 4410098718344

Option 3 ID : 4410098718342

Option 4 ID : 4410098718345

Status : Answered

Chosen Option : B

Q.24 Which European power was the first to reach India by the sea route, led by Vasco da Gama in 1498?

- Ans**
- ☒ A. British
- ☒ B. Portuguese
- ☒ C. Dutch
- ☒ D. French

Question ID : 4410092215552

Option 1 ID : 4410098718348

Option 2 ID : 4410098718346

Option 3 ID : 4410098718347

Option 4 ID : 4410098718349

Status : Answered

Chosen Option : B

Q.25 As per the Ministry of Ports, Shipping and Waterways, how many total Major Ports are there in India (As of June 2026)?

- Ans**
- ☒ A. 13
- ☒ B. 12
- ☒ C. 10
- ☒ D. 14

Question ID : 4410092215545

Option 1 ID : 4410098718321

Option 2 ID : 4410098718320

Option 3 ID : 4410098718318

Option 4 ID : 4410098718319

Status : Answered

Chosen Option : A

Section : Part 2 Numerical Ability

Q.1	Simplify: $\sqrt[3]{\frac{4096 \times 512}{64}}$
Ans	☒ A. 24
	☒ B. 48
	☒ C. 16
	☒ D. 32

Question ID : 4410092200471

Option 1 ID : 4410098659057

Option 2 ID : 4410098659059

Option 3 ID : 4410098659056

Option 4 ID : 4410098659058

Status : Answered

Chosen Option : D

Q.2	The average of four numbers is 20. If one of the numbers is 10, which equation represents the sum of the remaining three numbers?
Ans	☒ A. Sum of remaining three numbers = 10×3
	☒ B. Sum of remaining three numbers = $(20 - 10) \times 4$
	☒ C. Sum of remaining three numbers = $(20 + 10) \times 3$
	☒ D. Sum of remaining three numbers = $(20 \times 4) - 10$

Question ID : 4410092200456

Option 1 ID : 4410098658997

Option 2 ID : 4410098658998

Option 3 ID : 4410098658996

Option 4 ID : 4410098658999

Status : Answered

Chosen Option : D

Q.3 The sum of the present ages of Tommy and Arthur is 23 years. Four years ago from now, Arthur's age was half of Tommy's age at that time. What is Tommy's present age (in years)?

- Ans**
- ☐ A. 9
- ☐ B. 22
- ☐ C. 24
- ☒ D. 14

Question ID : 4410091220102

Option 1 ID : 4410094813834

Option 2 ID : 4410094813833

Option 3 ID : 4410094813832

Option 4 ID : 4410094813831

Status : Answered

Chosen Option : D

Q.4 A shopkeeper buys a school bag for ₹800 and marks it at 25% above the cost price. He sells it at the marked price. What is his profit?

- Ans**
- ☒ A. ₹200
- ☐ B. ₹150
- ☐ C. ₹180
- ☐ D. ₹250

Question ID : 4410092200477

Option 1 ID : 4410098659082

Option 2 ID : 4410098659080

Option 3 ID : 4410098659081

Option 4 ID : 4410098659083

Status : Answered

Chosen Option : A

Q.5 The cost of a washing machine is 15% less than the cost of a television. If the cost of the washing machine increases by 83% and that of the TV decreases by 40%, then what is the percentage change in the total cost of 5 washing machines and 6 televisions?

Ans ☒ A. Increase by 11%

☐ B. Decrease by 10%

☐ C. Decrease by 6%

☐ D. Increase by 9%

Question ID : 441009560862

Option 1 ID : 4410092198715

Option 2 ID : 4410092198716

Option 3 ID : 4410092198718

Option 4 ID : 4410092198717

Status : Answered

Chosen Option : A

Q.6 A jeep covers a distance of 100 km at a uniform speed. If the speed of the jeep is increased by 5 kmph, then it takes 1 hour less to cover the same distance. The original speed of the jeep is

Ans ☐ A. 18 kmph

☐ B. 23 kmph

☒ C. 20 kmph

☐ D. 21 kmph

Question ID : 441009371783

Option 1 ID : 4410091453050

Option 2 ID : 4410091453051

Option 3 ID : 4410091453048

Option 4 ID : 4410091453049

Status : Answered

Chosen Option : C

Q.7 An engineering company is designing a solid right prism whose base is a right-angled triangle with sides 9 cm, 12 cm, and 15 cm. The prism has a height of 20 cm. What is the total surface area of the prism?

Ans ☒ A. 792 cm²

☒ B. 828 cm²

☒ C. 810 cm²

☒ D. 846 cm²

Question ID : 4410092200506

Option 1 ID : 4410098659196

Option 2 ID : 4410098659198

Option 3 ID : 4410098659197

Option 4 ID : 4410098659199

Status : Answered

Chosen Option : B

Q.8 A box contains 5 red, 4 blue, and 3 green balls. Two balls are drawn one after another without replacement. What is the probability that the two balls are of different colours?

Ans ☒ A. $\frac{37}{66}$

☒ B. $\frac{41}{66}$

☒ C. $\frac{43}{66}$

☒ D. $\frac{47}{66}$

Question ID : 4410092200498

Option 1 ID : 4410098659164

Option 2 ID : 4410098659165

Option 3 ID : 4410098659166

Option 4 ID : 4410098659167

Status : Answered

Chosen Option : D

Q.9 A solid hemisphere of radius 28 cm is melted and recast into identical solid spheres, each of radius 7 cm. How many spheres can be formed?

Ans ☒ A. 28

☒ B. 36

☒ C. 24

☒ D. 32

Question ID : 4410092200501

Option 1 ID : 4410098659177

Option 2 ID : 4410098659179

Option 3 ID : 4410098659176

Option 4 ID : 4410098659178

Status : Answered

Chosen Option : D

Q.10 Portfolio A has stocks and bonds in the ratio 7 : 5, and Portfolio B has stocks and bonds in the ratio 9 : 7. If both portfolios have the same total investment, which portfolio has a larger share of stocks, and by what fraction of the total investment?

Ans ☒ A. Portfolio B by $\frac{1}{24}$

☒ B. Portfolio A by $\frac{1}{24}$

☒ C. Portfolio B by $\frac{1}{48}$

☒ D. Portfolio A by $\frac{1}{48}$

Question ID : 4410092200486

Option 1 ID : 4410098659119

Option 2 ID : 4410098659117

Option 3 ID : 4410098659118

Option 4 ID : 4410098659116

Status : Answered

Chosen Option : D

Q.11 If ₹10.75 is split equally among 5 people, how much does each person get?

Ans ☒ A. ₹2.35

☒ B. ₹2.15

☒ C. ₹2.25

☒ D. ₹2.05

Question ID : 4410092200465

Option 1 ID : 4410098659035

Option 2 ID : 4410098659033

Option 3 ID : 4410098659034

Option 4 ID : 4410098659032

Status : Answered

Chosen Option : B

Q.12 In a division sum, the divisor is 16 times the quotient and 5 times the remainder. If the remainder is 64, what is the dividend?

Ans ☒ A. 6468

☒ B. 6465

☒ C. 6469

☒ D. 6464

Question ID : 441009559907

Option 1 ID : 4410092194896

Option 2 ID : 4410092194898

Option 3 ID : 4410092194897

Option 4 ID : 4410092194895

Status : Answered

Chosen Option : C

Q.13 A sum of ₹8,000 is invested at a compound interest rate of 5% per annum for 2 years, interest compounded annually. Find the total amount at the end of 2 years.

Ans ☒ A. ₹9,020

☒ B. ₹8,420

☒ C. ₹8,820

☒ D. ₹8,620

Question ID : 4410092200462

Option 1 ID : 4410098659023

Option 2 ID : 4410098659020

Option 3 ID : 4410098659022

Option 4 ID : 4410098659021

Status : Answered

Chosen Option : C

Q.14 A charity divides its total funds between two projects. Project A receives 45% of the total funds, and Project B receives the remaining 55%. What is the ratio of the funds allocated to Project A and Project B?

Ans ☒ A. 9 : 10

☒ B. 11 : 9

☒ C. 5 : 6

☒ D. 9 : 11

Question ID : 4410092200474

Option 1 ID : 4410098659069

Option 2 ID : 4410098659071

Option 3 ID : 4410098659068

Option 4 ID : 4410098659070

Status : Answered

Chosen Option : D

Q.15 A shopkeeper buys sugar for ₹40 per kg using an accurate weighing scale. He sells it for ₹50 per kg but gives only 800 g instead of 1 kg. What is his actual profit percentage?

Ans ☒ A. 60%

☒ B. 56.25%

☒ C. 50%

☒ D. 62.5%

Question ID : 4410092200480

Option 1 ID : 4410098659094

Option 2 ID : 4410098659093

Option 3 ID : 4410098659092

Option 4 ID : 4410098659095

Status : Answered

Chosen Option : B

Q.16 A sum, when invested at 20% simple interest per annum, amounts to ₹8,400 after 2 years. What is the simple interest (in ₹) on the same sum at the same rate in 2 years?

Ans ☒ A. 2,400

☒ B. 4,800

☒ C. 9,600

☒ D. 1,200

Question ID : 441009568329

Option 1 ID : 4410092228436

Option 2 ID : 4410092228437

Option 3 ID : 4410092228439

Option 4 ID : 4410092228438

Status : Answered

Chosen Option : B

Q.17 A train covers a certain distance at a uniform speed. If the train had been 6 km/hr faster, it would have taken 4 hours less than the scheduled time. If the train was slower by 6 km/hr, it would have taken 6 hours more than the scheduled time. What is the length of the journey?

Ans ☒ A. 680 km

☒ B. 750 km

☒ C. 720 km

☒ D. 600 km

Question ID : 4410091560813

Option 1 ID : 4410096165451

Option 2 ID : 4410096165453

Option 3 ID : 4410096165452

Option 4 ID : 4410096165450

Status : Answered

Chosen Option : A

Q.18 A pharmaceutical company designs a spherical medicine capsule container with an internal radius of 6 cm. The container is to be completely filled with a liquid medicine. What is the volume of the container? (Use $\pi = 22/7$)

- Ans**
- ☒ A. $1223\frac{1}{3} \text{ cm}^3$
- ☒ B. $1121\frac{1}{7} \text{ cm}^3$
- ☒ C. $905\frac{1}{7} \text{ cm}^3$
- ☒ D. $1056\frac{1}{3} \text{ cm}^3$

Question ID : 4410092200509

Option 1 ID : 4410098659211

Option 2 ID : 4410098659210

Option 3 ID : 4410098659208

Option 4 ID : 4410098659209

Status : Answered

Chosen Option : C

Q.19 The weighted average of two exams is 70. Exam 1 counts twice as much as Exam 2. If the average score on Exam 2 is 60, what is the average score on Exam 1?

- Ans**
- ☒ A. 80
- ☒ B. 75
- ☒ C. 65
- ☒ D. 70

Question ID : 4410092200459

Option 1 ID : 4410098659011

Option 2 ID : 4410098659010

Option 3 ID : 4410098659008

Option 4 ID : 4410098659009

Status : Answered

Chosen Option : B

Q.20 A laboratory prepares a chemical solution by mixing Chemical A and Chemical B in the proportion 5 : 8. If 104 litres of the solution are prepared, how much quantity of Chemical B is required?

Ans ☒ A. 64 litres

☐ B. 72 litres

☐ C. 40 litres

☐ D. 36 litres

Question ID : 4410092200483

Option 1 ID : 4410098659106

Option 2 ID : 4410098659107

Option 3 ID : 4410098659105

Option 4 ID : 4410098659104

Status : Answered

Chosen Option : B

Q.21 The geometric mean of four positive numbers is 12. Three of the numbers are 6, 18, and 24. What is the fourth number?

Ans ☐ A. 10

☒ B. 8

☐ C. 4

☐ D. 6

Question ID : 4410092200492

Option 1 ID : 4410098659143

Option 2 ID : 4410098659142

Option 3 ID : 4410098659140

Option 4 ID : 4410098659141

Status : Answered

Chosen Option : A

Q.22 A solid metallic sphere of radius 6 cm is melted and recast into identical solid cones, each having a radius of 3 cm and a height of 12 cm. Assuming no loss of material, how many cones can be formed?

Ans ☐ A. 6

☐ B. 10

☒ C. 8

☐ D. 4

Question ID : 4410092200503

Option 1 ID : 4410098659185

Option 2 ID : 4410098659187

Option 3 ID : 4410098659186

Option 4 ID : 4410098659184

Status : Answered

Chosen Option : A

Q.23 Three research assistants, A, B, and C, have efficiencies in the ratio 3 : 4 : 5. Together, they can complete a project in 12 days. If A works alone for the first 8 days, after which B and C join and all three continue working together, how many more days are required to complete the remaining work?

Ans ☒ A. 12

☒ B. 6

☒ C. 8

☒ D. 10

Question ID : 4410092200495

Option 1 ID : 4410098659155

Option 2 ID : 4410098659152

Option 3 ID : 4410098659153

Option 4 ID : 4410098659154

Status : Answered

Chosen Option : C

Q.24 The amount A accumulated on an investment varies directly as the principal P and inversely as the square of the time t. When ₹20,000 is invested for 2 years, the accumulated amount is ₹25,000. If ₹45,000 is invested for 3 years under the same conditions, what will be the accumulated amount?

Ans ☒ A. ₹20,000

☒ B. ₹25,000

☒ C. ₹22,500

☒ D. ₹27,500

Question ID : 4410092200489

Option 1 ID : 4410098659128

Option 2 ID : 4410098659130

Option 3 ID : 4410098659129

Option 4 ID : 4410098659131

Status : Answered

Chosen Option : D

Q.25 A researcher schedules three periodic tasks: Alpha every 22 days, Beta every 36 days, and Gamma every 45 days. All three tasks are completed together today. She wants to schedule the next review meeting on the earliest day when all three tasks are due again. After how many days should she hold the review meeting?

Ans ☒ A. 1,980

☐ B. 990

☐ C. 1,320

☐ D. 1,650

Question ID : 4410092200468

Option 1 ID : 4410098659047

Option 2 ID : 4410098659044

Option 3 ID : 4410098659045

Option 4 ID : 4410098659046

Status : Answered

Chosen Option : C

Section : Part 3 Reasoning

Q.1 If 2 is added to each odd digit and 3 is subtracted from each even digit in the number 1467538, how many digits will appear more than once in the new number thus formed?

Ans ☐ A. Three

☐ B. None

☐ C. One

☒ D. Two

Question ID : 441009550870

Option 1 ID : 4410092158797

Option 2 ID : 4410092158794

Option 3 ID : 4410092158795

Option 4 ID : 4410092158796

Status : Answered

Chosen Option : D

Q.2 HVJC is related to IXIA in a certain way based on the English alphabetical order. In the same way, XQDI is related to YSCG. To which of the following is RWPM related, following the same logic?

Ans ☒ A. SYOK

☐ B. SXPK

☐ C. SYPJ

☐ D. RXOK

Question ID : 441009805366

Option 1 ID : 4410093175588

Option 2 ID : 4410093175590

Option 3 ID : 4410093175589

Option 4 ID : 4410093175587

Status : Answered

Chosen Option : A

Q.3 An accurate clock shows 2:00 PM. By how many degrees will the hour hand rotate when the clock shows 11:00 PM?

Ans ☒ A. 270°

☐ B. 210°

☐ C. 240°

☐ D. 180°

Question ID : 4410092227684

Option 1 ID : 4410098765124

Option 2 ID : 4410098765126

Option 3 ID : 4410098765125

Option 4 ID : 4410098765127

Status : Answered

Chosen Option : C

Q.4 Riya starts walking from a point P. She walks towards east for 2 m. Then she takes a right turn and walks for 3 m. Then she takes left turn and walks for 7 m. Then she takes a right turn and walks for 12 m. Then she takes a right turn and walks for 4 m. Again, she takes a right turn and walks for 5 m. Then she takes a left turn and walks for 2 m. Then she takes a left turn and walks for 6 m. Then she takes a right turn and walks for 3 m. At last, she takes a right turn and walks for 5 m before stopping at a point Q. Towards which direction is point P located with respect to point Q? (All turns are 90-degree turns only unless specified.)

- Ans**
- ☐ A. South
- ☐ B. North East
- ☒ C. North
- ☐ D. South West

Question ID : 44100964618

Option 1 ID : 441009257441

Option 2 ID : 441009257442

Option 3 ID : 441009257443

Option 4 ID : 441009257444

Status : Answered

Chosen Option : C

Q.5 If 21 October 2003 is a Tuesday, then what will be the day of the week on 17 June 2011?

- Ans**
- ☐ A. Thursday
- ☒ B. Friday
- ☐ C. Saturday
- ☐ D. Sunday

Question ID : 441009723298

Option 1 ID : 4410092846944

Option 2 ID : 4410092846945

Option 3 ID : 4410092846946

Option 4 ID : 4410092846947

Status : Answered

Chosen Option : C

Q.6 22 is related to 157 following a certain logic. Following the same logic, 34 is related to 241. To which of the following is 45 related to, following the same logic?
(NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)

Ans ☒ A. 316

☒ B. 318

☒ C. 314

☒ D. 320

Question ID : 4410092226647

Option 1 ID : 4410098761874

Option 2 ID : 4410098761873

Option 3 ID : 4410098761875

Option 4 ID : 4410098761876

Status : Answered

Chosen Option : B

Q.7 Each consonant in the word RECORD is changed to the letter immediately following it in the English alphabetical order and vowels are kept unchanged. How many vowels are there in the word so formed?

Ans ☒ A. Two

☒ B. Five

☒ C. Three

☒ D. Four

Question ID : 441009790861

Option 1 ID : 4410093117444

Option 2 ID : 4410093117443

Option 3 ID : 4410093117445

Option 4 ID : 4410093117446

Status : Answered

Chosen Option : C

Q.8 What should come in place of the question mark (?) in the following series based on the English alphabetical order?

LOJ MPK NQL ORM ?

Ans ☒ A. SPM

☒ B. PSN

☒ C. SPN

☒ D. PSM

Question ID : 441009732344

Option 1 ID : 4410092883131

Option 2 ID : 4410092883128

Option 3 ID : 4410092883130

Option 4 ID : 4410092883129

Status : Answered

Chosen Option : D

Q.9 Based on the English alphabetical order, three of the following four letter-clusters are alike in a certain way and thus form a group. Which letter-cluster DOES NOT belong to that group?

(Note: The odd one out is not based on the number of consonants/vowels or their position in the letter-cluster.)

Ans ☒ A. AHD

☒ B. LSP

☒ C. QXT

☒ D. NUQ

Question ID : 441009385010

Option 1 ID : 4410091505665

Option 2 ID : 4410091505662

Option 3 ID : 4410091505664

Option 4 ID : 4410091505663

Status : Answered

Chosen Option : A

Q.10 In this question, a statement is followed by two courses of action, numbered I and II. You must assume everything in the statement to be true, and on the basis of the information given in the statement, decide which of the courses of action logically follow(s) for pursuing.

Statement:

Frequent breakdown of city buses in City R has caused inconvenience to daily commuters.

Courses of Action:

I. Older buses should be repaired and maintained regularly to avoid breakdowns.

II. Taxes on private vehicles should be increased.

Ans ☒ A. Neither I nor II follows

☒ B. Only I follows

☒ C. Both I and II follow

☒ D. Only II follows

Question ID : 4410091207078

Option 1 ID : 4410094761841

Option 2 ID : 4410094761838

Option 3 ID : 4410094761840

Option 4 ID : 4410094761839

Status : Answered

Chosen Option : B

Q.11 Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statements.

Statements:

All plates are spoons.

All spoons are bowls.

Conclusions:

(I) Some bowls are plates.

(II) Some bowls are spoons.

Ans ☒ A. Only conclusion (II) follows.

☒ B. Neither conclusion (I) nor (II) follows.

☒ C. Both conclusions (I) and (II) follow.

☒ D. Only conclusion (I) follows.

Question ID : 44100928285

Option 1 ID : 441009112673

Option 2 ID : 441009112674

Option 3 ID : 441009112675

Option 4 ID : 441009112672

Status : Answered

Chosen Option : C

Q.12 In this question, a question is followed by two statements numbered (I) and (II). You have to decide whether the data provided in the statements are sufficient to answer the question. Read both the statements and decide the appropriate answer.
 Five innovation centres – Apex, Bloom, Crest, Drive and Elevate – are allotted Building Numbers 1 to 5, one distinct building to each centre. Which building is allotted to Crest?
 (I) Crest is allotted a higher-numbered building than Bloom but a lower-numbered building than Drive.
 (II) Apex is allotted Building 1, Bloom is allotted Building 2, and Elevate is allotted Building 5.

- Ans** ☒ A. Both statements I and II together (and not statement I alone or statement II alone) are sufficient to answer the question
- ☐ B. Both statements I and II together are NOT sufficient to answer the question
- ☐ C. Data in Statement II alone is sufficient to answer the question while data in Statement I is not
- ☐ D. Data in Statement I alone is sufficient to answer the question while data in Statement II is not

Question ID : 4410092226641

Option 1 ID : 4410098761849

Option 2 ID : 4410098761852

Option 3 ID : 4410098761851

Option 4 ID : 4410098761850

Status : Answered

Chosen Option : A

Q.13 M, N, O, P, W, X and Y are sitting around a circular table, facing the centre of the table.
 P sits to the immediate left of N. W sits to the immediate left of M. M sits second to the left of X. O sits third to the left of N.
 Who sits third to the right of Y?

- Ans** ☒ A. O
- ☐ B. M
- ☐ C. Y
- ☐ D. P

Question ID : 441009810004

Option 1 ID : 4410093194140

Option 2 ID : 4410093194139

Option 3 ID : 4410093194142

Option 4 ID : 4410093194141

Status : Answered

Chosen Option : A

Q.14 What should come in place of the question mark (?) in the given series?

68 79 92 107 124 ?

Ans ☒ A. 143

☐ B. 144

☐ C. 145

☐ D. 142

Question ID : 441009338566

Option 1 ID : 4410091320228

Option 2 ID : 4410091320230

Option 3 ID : 4410091320231

Option 4 ID : 4410091320229

Status : Answered

Chosen Option : A

Q.15 What should come in place of '?' in the given series?

54 60 72 90 114 ?

Ans ☐ A. 147

☐ B. 142

☒ C. 144

☐ D. 146

Question ID : 4410091201851

Option 1 ID : 4410094741189

Option 2 ID : 4410094741186

Option 3 ID : 4410094741187

Option 4 ID : 4410094741188

Status : Answered

Chosen Option : C

Q.16 C, D, E, F, Y and Z live on six different floors of the same building. The lowermost floor in the building is numbered 1, the floor above it is numbered 2 and so on, till the topmost floor, which is numbered 6. E lives on floor number 4. Only two people live between E and Y. Only D lives between E and F. C lives immediately below E. Who lives on floor number 2?

Ans ☒ A. C

☒ B. D

☒ C. F

☒ D. Z

Question ID : 441009693452

Option 1 ID : 4410092727840

Option 2 ID : 4410092727839

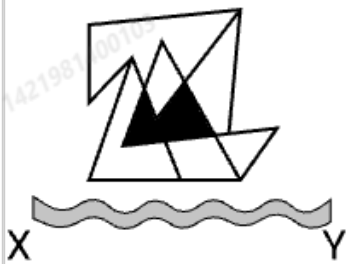
Option 3 ID : 4410092727838

Option 4 ID : 4410092727841

Status : Answered

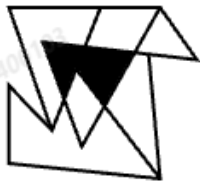
Chosen Option : D

Q.17 If the water source is placed along X-Y, as shown in the figure, which of the given options will be the water reflection of the given figure.

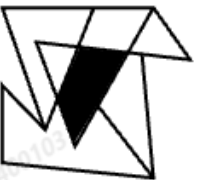


Ans

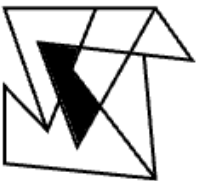
✓ A.



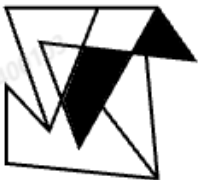
✗ B.



✗ C.



✗ D.



Question ID : 4410092228111

Option 1 ID : 4410098766769

Option 2 ID : 4410098766768

Option 3 ID : 4410098766770

Option 4 ID : 4410098766771

Status : Answered

Chosen Option : A

Q.18 Refer to the given letter and symbol series and answer the question that follows. Counting to be done from left to right only.

(Left) N F K * Q % ? U H # J O \$ * H D @ Z + J V (Right)

How many such symbols are there, each of which is immediately preceded by a letter and also immediately followed by a letter?

Ans ☒ A. Four

☐ B. One

☐ C. Two

☐ D. Three

Question ID : 441009782646

Option 1 ID : 4410093084590

Option 2 ID : 4410093084587

Option 3 ID : 4410093084588

Option 4 ID : 4410093084589

Status : Answered

Chosen Option : A

Q.19 In a certain code,
'blanket pillow mattress' is coded as 'vx kp rd'
'mattress curtain carpet' is coded as 'rd jm ty'
'carpet blanket cushion' is coded as 'vx nq jm'
(all the codes are two-letter codes only)
Which word is coded as 'ty'?

Ans ☐ A. pillow

☒ B. curtain

☐ C. mattress

☐ D. carpet

Question ID : 4410092226637

Option 1 ID : 4410098761836

Option 2 ID : 4410098761833

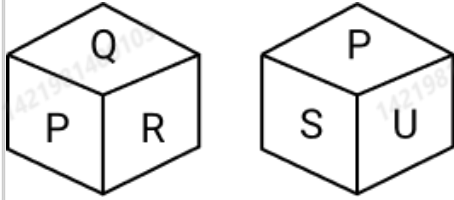
Option 3 ID : 4410098761835

Option 4 ID : 4410098761834

Status : Answered

Chosen Option : B

- Q.20** Six letters, P, Q, R, S, T and U, are written on different faces of a dice. Two positions of this dice are shown in the figure. Find the letter on the face opposite to T.



Ans ☒ A. U

☒ B. Q

☒ C. S

☒ D. P

Question ID : 441009865742

Option 1 ID : 4410093416091

Option 2 ID : 4410093416090

Option 3 ID : 4410093416088

Option 4 ID : 4410093416089

Status : Answered

Chosen Option : A

- Q.21** Seven boxes, C, H, A, I, R, E and D, are kept one over the other but not necessarily in the same order. Only H is kept above R. Only two boxes are kept between R and E. Only A is kept below I. D is not kept immediately above E. How many boxes are kept between C and A?

Ans ☒ A. Four

☒ B. One

☒ C. Three

☒ D. Two

Question ID : 4410091526437

Option 1 ID : 4410096028865

Option 2 ID : 4410096028862

Option 3 ID : 4410096028864

Option 4 ID : 4410096028863

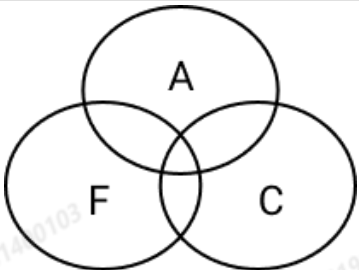
Status : Answered

Chosen Option : B

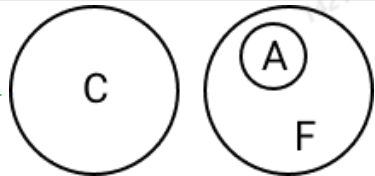
Q.22 Select the Venn diagram that best illustrates the relationship between the following classes
Carrot, Fruit, Apple

Ans

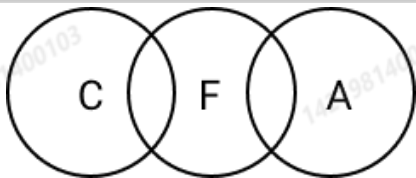
☒ A.



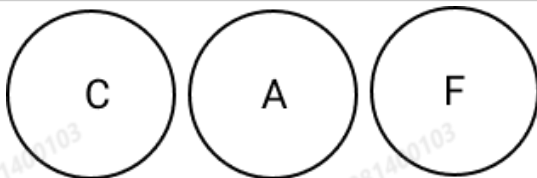
☒ B.



☒ C.



☒ D.



Question ID : 4410092226645

Option 1 ID : 4410098761868

Option 2 ID : 4410098761867

Option 3 ID : 4410098761865

Option 4 ID : 4410098761866

Status : Answered

Chosen Option : D

Q.23 Meena is the mother of Anjali. Anjali is the wife of Sandeep. Sandeep is the father of Rohan. Rohan is the brother of Nisha. How is Meena related to Nisha?

Ans ☒ A. Mother's mother

☐ B. Father's mother

☐ C. Father's sister

☐ D. Mother's sister

Question ID : 4410092226633

Option 1 ID : 4410098761817

Option 2 ID : 4410098761819

Option 3 ID : 4410098761820

Option 4 ID : 4410098761818

Status : Answered

Chosen Option : A

Q.24 Refer to the given letter and symbol series and answer the question that follows. Counting to be done from left to right only.

(Left) % J \$ K Y © £ £ X N A \$ C & % B © B O € D P (Right)

How many such letters are there, each of which is immediately preceded by a symbol and also immediately followed by a symbol?

Ans ☐ A. 5

☐ B. 2

☐ C. 4

☒ D. 3

Question ID : 441009695654

Option 1 ID : 4410092736649

Option 2 ID : 4410092736648

Option 3 ID : 4410092736647

Option 4 ID : 4410092736646

Status : Answered

Chosen Option : D

Q.25 C, O, R, P and E have different heights. Only three people have heights between E and R. P is taller than O but shorter than C. E is taller than R. How many people are shorter than P?

- Ans**
- ☒ A. One
- ☒ B. Two
- ☒ C. Four
- ☒ D. Three

Question ID : 4410091646975

Option 1 ID : 4410096504161

Option 2 ID : 4410096504162

Option 3 ID : 4410096504164

Option 4 ID : 4410096504163

Status : Answered

Chosen Option : B

Section : Part 4 General English

Q.1 Identify the error in the given sentence and choose the option that is the rectified version.
This time next Saturday, I will relax in my new house.

- Ans**
- ☒ A. This time next Saturday, I am relaxing in my new house.
- ☒ B. This time next Saturday, I relax in my new house.
- ☒ C. This time next Saturday, I will have relaxed in my new house.
- ☒ D. This time next Saturday, I will be relaxing in my new house.

Question ID : 4410092302002

Option 1 ID : 4410099053723

Option 2 ID : 4410099053724

Option 3 ID : 4410099053725

Option 4 ID : 4410099053726

Status : Answered

Chosen Option : D

Q.2 Choose the most appropriate option from the given choices to correctly complete the sentence.

The research paper was praised for its originality, but it was criticized because _____ its theoretical framework _____ its empirical analysis met the expected standards.

- Ans**
- ☒ A. either; or
- ☒ B. neither; or
- ☒ C. neither; nor
- ☒ D. not only; but also

Question ID : 4410091560963

Option 1 ID : 4410096166063

Option 2 ID : 4410096166064

Option 3 ID : 4410096166062

Option 4 ID : 4410096166065

Status : Answered

Chosen Option : C

Q.3 Select the most appropriate option to fill in the blanks. If no article is required, select the option with 'No article required'.

They were looking for _____ 17-year-old girl wearing _____ pair of blue shoes.

- Ans**
- ☒ A. a; No article required
- ☒ B. No article required; an
- ☒ C. an; a
- ☒ D. a; a

Question ID : 441009126998

Option 1 ID : 441009504681

Option 2 ID : 441009504680

Option 3 ID : 441009504679

Option 4 ID : 441009504678

Status : Answered

Chosen Option : D

Q.4 Select the most appropriate option to fill in the blanks. If no article is required, select the option with 'No article required'.

The symposium opened with ___ incisive critique of postcolonial theory and concluded with ___ ambitious rethinking of ecological ethics.

Ans ☒ A. No article required; No article required

☒ B. a; an

☒ C. an; an

☒ D. an; No article required

Question ID : 4410091292000

Option 1 ID : 4410095100083

Option 2 ID : 4410095100080

Option 3 ID : 4410095100081

Option 4 ID : 4410095100082

Status : Answered

Chosen Option : A

Q.5 Select the most appropriate meaning of the given idiom.

Herculean task

Ans ☒ A. a task or a puzzle that is extremely confusing

☒ B. a difficult task that requires a huge amount of effort

☒ C. a task with an urgent deadline

☒ D. a task that is fun and enjoyable

Question ID : 441009230312

Option 1 ID : 441009900071

Option 2 ID : 441009900072

Option 3 ID : 441009900073

Option 4 ID : 441009900074

Status : Answered

Chosen Option : B

Q.6 Select the option where the given sentence has been correctly converted to the Passive Voice.
Submit the expense report by Friday.

- Ans**
- ☒ A. You are submitting the expense report by Friday.
 - ☒ B. You should submit the expense report by Friday.
 - ☒ C. The expense report has to be submitted by Friday.
 - ☒ D. Let the expense report be submitted by Friday.

Question ID : 4410092306002

Option 1 ID : 4410099069575

Option 2 ID : 4410099069573

Option 3 ID : 4410099069574

Option 4 ID : 4410099069572

Status : Answered

Chosen Option : D

Q.7 Select the option where the given sentence has been correctly converted to the Active Voice:
Is the report being prepared ?

- Ans**
- ☒ A. Will they prepare the report?
 - ☒ B. Are they preparing the report?
 - ☒ C. Do they prepare the report?
 - ☒ D. Have they prepared the report?

Question ID : 4410092306033

Option 1 ID : 4410099069683

Option 2 ID : 4410099069681

Option 3 ID : 4410099069680

Option 4 ID : 4410099069682

Status : Answered

Chosen Option : B

Q.8 A sentence with a grammatical error is given here. Select the option that will rectify this error.

Her research differed with the established theory on several key points.

- Ans**
- ☒ A. Her research differed to the established theory on several key points.
- ☒ B. Her research differed against the established theory on several key points.
- ☒ C. Her research differed from the established theory on several key points.
- ☒ D. Her research differed than the established theory on several key points.

Question ID : 4410091648024

Option 1 ID : 4410096508362

Option 2 ID : 4410096508359

Option 3 ID : 4410096508361

Option 4 ID : 4410096508360

Status : Answered

Chosen Option : A

Q.9 Arrange the following to form a grammatically correct sentence:

P: had been overlooked in earlier analyses

Q: the subtle thematic patterns

R: a deeper layer of meaning revealed that

S: upon closer examination of the text

Ans ☒ A. S R Q P

☒ B. S Q R P

☒ C. Q R S P

☒ D. P S Q R

Question ID : 4410092220892

Option 1 ID : 4410098739214

Option 2 ID : 4410098739217

Option 3 ID : 4410098739215

Option 4 ID : 4410098739216

Status : Answered

Chosen Option : A

Q.10 From the fragments below, which order most coherently expresses a logical sequence?-

- (A) despite the preliminary evidence supporting the original framework
 (B) necessitating a series of methodological revisions
 (C) the experimental design was subsequently modified
 (D) when unforeseen variables began to influence the reliability of the findings

Ans ☒ A. A, D, C, B

☒ B. A, C, B, D

☒ C. A, B, C, D

☒ D. A, C, D, B,

Question ID : 4410092220888

Option 1 ID : 4410098739201

Option 2 ID : 4410098739199

Option 3 ID : 4410098739200

Option 4 ID : 4410098739198

Status : Answered

Chosen Option : B

Q.11 Select the best completion for the indirect speech conversion of the given sentence.

She asked, 'Why did you not attend the seminar last Friday?'
 She asked why _____.

Ans ☒ A. I did not attend the seminar last Friday

☒ B. I had not attended the seminar the previous Friday

☒ C. I have not attended the seminar that Friday

☒ D. I do not attend the seminar the previous Friday

Question ID : 4410092220876

Option 1 ID : 4410098739151

Option 2 ID : 4410098739153

Option 3 ID : 4410098739152

Option 4 ID : 4410098739150

Status : Answered

Chosen Option : B

Q.12 Select the wrongly spelt word from the alternatives given below.

- Ans**
- ☒ A. Rabbinic
 - ☒ B. Arsenic
 - ☒ C. Opponancy
 - ☒ D. Exoneration

Question ID : 4410094058

Option 1 ID : 44100916151

Option 2 ID : 44100916154

Option 3 ID : 44100916152

Option 4 ID : 44100916153

Status : Answered

Chosen Option : D

Q.13 Identify the error in the given sentence and choose the option that is the rectified version.
The train is arriving at platform 3 at 5 PM sharp, according to the printed timetable.

- Ans**
- ☒ A. The train has arrived at platform 3 at 5 PM sharp, according to the printed timetable.
 - ☒ B. The train arrived at platform 3 at 5 PM sharp, according to the printed timetable.
 - ☒ C. The train has been arriving at platform 3 at 5 PM sharp, according to the printed timetable.
 - ☒ D. The train arrives at platform 3 at 5 PM sharp, according to the printed timetable.

Question ID : 4410092301999

Option 1 ID : 4410099053712

Option 2 ID : 4410099053711

Option 3 ID : 4410099053714

Option 4 ID : 4410099053713

Status : Answered

Chosen Option : D

Q.14 Select the option that completes the sentence appropriately.

Ravi said that he _____ that book ____.

- Ans**
- ☒ A. was reading / now
 - ☒ B. had been reading / now
 - ☒ C. had read / now
 - ☒ D. was reading / then

Question ID : 4410091311996

Option 1 ID : 4410095179664

Option 2 ID : 4410095179667

Option 3 ID : 4410095179666

Option 4 ID : 4410095179665

Status : Answered

Chosen Option : D

Q.15 Select the one word that substitutes for the underlined words:
Meera liked Krishna because she found him to be 'A person who is skilled in several arts.'

- Ans**
- ☒ A. Dilettante
 - ☒ B. Specialist
 - ☒ C. Polymath
 - ☒ D. Novice

Question ID : 4410092220884

Option 1 ID : 4410098739185

Option 2 ID : 4410098739184

Option 3 ID : 4410098739183

Option 4 ID : 4410098739182

Status : Answered

Chosen Option : A

Q.16 Select the wrongly spelt word.

- Ans**
- ☒ A. Acquiesence
 - ☒ B. Aesthetic
 - ☒ C. Conscientious
 - ☒ D. Magnanimous

Question ID : 4410091305621

Option 1 ID : 4410095154563

Option 2 ID : 4410095154562

Option 3 ID : 4410095154561

Option 4 ID : 4410095154564

Status : Answered

Chosen Option : C

Q.17 Choose the most appropriate synonym of the underlined word from the options given below.

The minister's speech was filled with bombastic language that lacked real substance.

- Ans**
- ☒ A. Queer
 - ☒ B. Rhapsodic
 - ☒ C. Quiet
 - ☒ D. Zealous

Question ID : 441009289976

Option 1 ID : 4410091126853

Option 2 ID : 4410091126854

Option 3 ID : 4410091126852

Option 4 ID : 4410091126855

Status : Answered

Chosen Option : B

Q.18 Choose the correct one-word for the group of words given below.-
a fear or dislike of foreign people, customs, or cultures

- Ans**
- ☒ A. Xenophobia
 - ☒ B. Misanthropy
 - ☒ C. Philology
 - ☒ D. Cynophobia

Question ID : 4410092220887

Option 1 ID : 4410098739194

Option 2 ID : 4410098739196

Option 3 ID : 4410098739197

Option 4 ID : 4410098739195

Status : Answered

Chosen Option : D

Q.19 Select the option that rectifies the error in the given sentence.

By the time the lecture began, most students have already found their seats.

- Ans**
- ☒ A. the lecture begin,
 - ☒ B. By the times
 - ☒ C. already find their seats.
 - ☒ D. most students had

Question ID : 4410091778496

Option 1 ID : 4410097013352

Option 2 ID : 4410097013351

Option 3 ID : 4410097013354

Option 4 ID : 4410097013353

Status : Answered

Chosen Option : D

Q.20 Choose the most appropriate option that has the same meaning as the given group of words:
to steal a march on someone

- Ans**
- ☒ A. To physically take something from someone while they are distracted during a parade or procession
 - ☒ B. To gain advantage over someone by acting earlier or more quickly than they do
 - ☒ C. To deliberately delay your own actions so that the other person makes the first mistake
 - ☒ D. To secretly follow someone and copy their strategy without their knowledge

Question ID : 4410092302779

Option 1 ID : 4410099056716

Option 2 ID : 4410099056718

Option 3 ID : 4410099056719

Option 4 ID : 4410099056717

Status : Answered

Chosen Option : D

Comprehension:

Read the given passage and answer the questions that follow.

The First Wave of Feminism emerged in the late 19th and early 20th centuries as a significant social and political movement aimed at securing basic legal rights for women. It was primarily focused on issues such as women's suffrage, property rights, and access to education. This movement developed in response to widespread gender inequality, where women were largely excluded from political participation and denied legal autonomy. One of the central goals of the movement was the right to vote, which became a powerful symbol of equality and citizenship. Activists organized campaigns, protests, and petitions to demand suffrage. In countries like the United States and the United Kingdom, the movement gained momentum through the efforts of prominent leaders such as Susan B. Anthony and Emmeline Pankhurst. Their advocacy helped bring women's issues into mainstream political discourse. The First Wave was largely driven by middle-class women and often focused on legal reforms rather than broader social change. While it achieved important victories—such as women gaining the right to vote in several countries—it did not fully address issues related to race, class, and working conditions. As a result, its impact was both transformative and limited. Despite its shortcomings, the First Wave laid the foundation for future feminist movements. It established the importance of collective action and highlighted the need for gender equality in public life. Its legacy continues to influence contemporary discussions on women's rights and social justice.

SubQuestion No : 21

Q.21 What is the central theme of the passage?

- Ans**
- ☒ A. The history of education systems during 19th and 20th centuries
 - ☒ B. The development and achievements of the First Wave of Feminism
 - ☒ C. Political systems in the 20th century feminist movement
 - ☒ D. The economic roles of women in different societies across the world

Question ID : 4410091872584

Option 1 ID : 4410097377952

Option 2 ID : 4410097377951

Option 3 ID : 4410097377953

Option 4 ID : 4410097377950

Status : Answered

Chosen Option : B

Comprehension:

Read the given passage and answer the questions that follow.

The First Wave of Feminism emerged in the late 19th and early 20th centuries as a significant social and political movement aimed at securing basic legal rights for women. It was primarily focused on issues such as women's suffrage, property rights, and access to education. This movement developed in response to widespread gender inequality, where women were largely excluded from political participation and denied legal autonomy. One of the central goals of the movement was the right to vote, which became a powerful symbol of equality and citizenship. Activists organized campaigns, protests, and petitions to demand suffrage. In countries like the United States and the United Kingdom, the movement gained momentum through the efforts of prominent leaders such as Susan B. Anthony and Emmeline Pankhurst. Their advocacy helped bring women's issues into mainstream political discourse. The First Wave was largely driven by middle-class women and often focused on legal reforms rather than broader social change. While it achieved important victories—such as women gaining the right to vote in several countries—it did not fully address issues related to race, class, and working conditions. As a result, its impact was both transformative and limited. Despite its shortcomings, the First Wave laid the foundation for future feminist movements. It established the importance of collective action and highlighted the need for gender equality in public life. Its legacy continues to influence contemporary discussions on women's rights and social justice.

SubQuestion No : 22

Q.22 What can be inferred about the First Wave of Feminism?

Ans ☒ A. It was led entirely by working-class women

☒ B. It focused mainly on legal and political rights

☒ C. It addressed all forms of inequality equally

☒ D. It ignored the issue of voting rights

Question ID : 4410091872585

Option 1 ID : 4410097377957

Option 2 ID : 4410097377956

Option 3 ID : 4410097377954

Option 4 ID : 4410097377955

Status : Answered

Chosen Option : B

Comprehension:

Read the given passage and answer the questions that follow.

The First Wave of Feminism emerged in the late 19th and early 20th centuries as a significant social and political movement aimed at securing basic legal rights for women. It was primarily focused on issues such as women's suffrage, property rights, and access to education. This movement developed in response to widespread gender inequality, where women were largely excluded from political participation and denied legal autonomy. One of the central goals of the movement was the right to vote, which became a powerful symbol of equality and citizenship. Activists organized campaigns, protests, and petitions to demand suffrage. In countries like the United States and the United Kingdom, the movement gained momentum through the efforts of prominent leaders such as Susan B. Anthony and Emmeline Pankhurst. Their advocacy helped bring women's issues into mainstream political discourse. The First Wave was largely driven by middle-class women and often focused on legal reforms rather than broader social change. While it achieved important victories—such as women gaining the right to vote in several countries—it did not fully address issues related to race, class, and working conditions. As a result, its impact was both transformative and limited. Despite its shortcomings, the First Wave laid the foundation for future feminist movements. It established the importance of collective action and highlighted the need for gender equality in public life. Its legacy continues to influence contemporary discussions on women's rights and social justice.

SubQuestion No : 23

Q.23 Which of the following is a fact mentioned in the passage?

- Ans**
- ☒ A. Women had full political rights before the movement even the right to vote
 - ☒ B. The First Wave ended all gender inequality across the world
 - ☒ C. The movement achieved voting rights for women in several countries
 - ☒ D. It was led only by men who believed in equality in the society

Question ID : **4410091872586**

Option 1 ID : **4410097377960**

Option 2 ID : **4410097377959**

Option 3 ID : **4410097377958**

Option 4 ID : **4410097377961**

Status : **Answered**

Chosen Option : **C**

Comprehension:

Read the given passage and answer the questions that follow.

The First Wave of Feminism emerged in the late 19th and early 20th centuries as a significant social and political movement aimed at securing basic legal rights for women. It was primarily focused on issues such as women's suffrage, property rights, and access to education. This movement developed in response to widespread gender inequality, where women were largely excluded from political participation and denied legal autonomy. One of the central goals of the movement was the right to vote, which became a powerful symbol of equality and citizenship. Activists organized campaigns, protests, and petitions to demand suffrage. In countries like the United States and the United Kingdom, the movement gained momentum through the efforts of prominent leaders such as Susan B. Anthony and Emmeline Pankhurst. Their advocacy helped bring women's issues into mainstream political discourse. The First Wave was largely driven by middle-class women and often focused on legal reforms rather than broader social change. While it achieved important victories—such as women gaining the right to vote in several countries—it did not fully address issues related to race, class, and working conditions. As a result, its impact was both transformative and limited. Despite its shortcomings, the First Wave laid the foundation for future feminist movements. It established the importance of collective action and highlighted the need for gender equality in public life. Its legacy continues to influence contemporary discussions on women's rights and social justice.

SubQuestion No : 24

Q.24 What is a limitation of the First Wave as discussed in the passage?

Ans ☒ A. It opposed education for women

☒ B. It ignored issues of race and class

☒ C. It failed to gain any legal rights

☒ D. It rejected political participation

Question ID : 4410091872587

Option 1 ID : 4410097377964

Option 2 ID : 4410097377963

Option 3 ID : 4410097377962

Option 4 ID : 4410097377965

Status : Answered

Chosen Option : B

Comprehension:

Read the given passage and answer the questions that follow.

The First Wave of Feminism emerged in the late 19th and early 20th centuries as a significant social and political movement aimed at securing basic legal rights for women. It was primarily focused on issues such as women's suffrage, property rights, and access to education. This movement developed in response to widespread gender inequality, where women were largely excluded from political participation and denied legal autonomy. One of the central goals of the movement was the right to vote, which became a powerful symbol of equality and citizenship. Activists organized campaigns, protests, and petitions to demand suffrage. In countries like the United States and the United Kingdom, the movement gained momentum through the efforts of prominent leaders such as Susan B. Anthony and Emmeline Pankhurst. Their advocacy helped bring women's issues into mainstream political discourse. The First Wave was largely driven by middle-class women and often focused on legal reforms rather than broader social change. While it achieved important victories—such as women gaining the right to vote in several countries—it did not fully address issues related to race, class, and working conditions. As a result, its impact was both transformative and limited. Despite its shortcomings, the First Wave laid the foundation for future feminist movements. It established the importance of collective action and highlighted the need for gender equality in public life. Its legacy continues to influence contemporary discussions on women's rights and social justice.

SubQuestion No : 25

Q.25 What is the structure of the passage?

- Ans**
- ☒ A. Cause–Effect–Solution
 - ☒ B. Chronological–Descriptive
 - ☒ C. Definition–Explanation–Evaluation
 - ☒ D. Problem–Solution–Conclusion

Question ID : 4410091872588

Option 1 ID : 4410097377968

Option 2 ID : 4410097377967

Option 3 ID : 4410097377966

Option 4 ID : 4410097377969

Status : Answered

Chosen Option : B

Section : Part 5 Domain Knowledge

Q.1 What does the read() system call in an operating system do?

- Ans**
- ☒ A. Blocks until the requested number of bytes is read
 - ☒ B. Copies data directly from device hardware into user space without kernel intervention
 - ☒ C. Runs entirely in user mode for performance
 - ☒ D. Copies data from kernel space to user space, possibly after reading from device buffers

Question ID : 4410091187728

Option 1 ID : 4410094685225

Option 2 ID : 4410094685222

Option 3 ID : 4410094685223

Option 4 ID : 4410094685224

Status : Answered

Chosen Option : B

Q.2 Which of the following correctly describes loopback addressing?

Ans ☒ A. Used for local LAN broadcast.

☒ B. Range from 127.0.0.0 to 127.255.255.255 used for internal host testing.

☒ C. Used to map private IP to public IP.

☒ D. Reserved for router multicast.

Question ID : 4410091441064

Option 1 ID : 4410095692922

Option 2 ID : 4410095692921

Option 3 ID : 4410095692924

Option 4 ID : 4410095692923

Status : Answered

Chosen Option : B

Q.3 Suppose an array contains N elements. Which of the following best represents the time complexity of traversing every alternate element?

Ans ☒ A. $O(N^2)$

☒ B. $O(\log N)$

☒ C. $O(N)$

☒ D. $O(1)$

Question ID : 4410092215471

Option 1 ID : 4410098718025

Option 2 ID : 4410098718024

Option 3 ID : 4410098718023

Option 4 ID : 4410098718022

Status : Answered

Chosen Option : C

Q.4 Consider an FSM that accepts all strings over {a, b} with an even number of a's. Which regular expression corresponds to it?

Ans ☒ A. $(a+b)^*a(a+b)^*(b+c)$

☒ B. $(b*ab^*)^*$

☒ C. $(b*(ab*ab^*))^*$

☒ D. $(a+b)^*a(a+b)^*(a+b)$

Question ID : 4410091210303

Option 1 ID : 4410094774717

Option 2 ID : 4410094774714

Option 3 ID : 4410094774715

Option 4 ID : 4410094774716

Status : Answered

Chosen Option : A

Q.5 Identify the correct option for project operation in relational algebra:
Statement-1: The project operation is a binary operation that returns its argument relation, with certain tuple left out.
Statement-2: The result of the PROJECT operation can be visualized as a horizontal partition of the relation into two relations: one has the needed rows (tuples) and contains the result of the operation, and the other contains the discarded rows.

- Ans**
- ☒ A. Statement-1 is True and Statement-2 is False
 - ☒ B. Statement-1 is True and Statement-2 is True
 - ☒ C. Statement-1 is False and Statement-2 is True
 - ☒ D. Statement-1 is False and Statement-2 is False

Question ID : 4410091443746

Option 1 ID : 4410095703545

Option 2 ID : 4410095703548

Option 3 ID : 4410095703546

Option 4 ID : 4410095703547

Status : Answered

Chosen Option : B

Q.6 In a circular queue using arrays, the rear is at index 4, front is at index 2, and capacity = 5. Which expression checks if the queue is full?

- Ans**
- ☒ A. $((\text{rear} + 1) \% \text{capacity} == \text{front})$
 - ☒ B. $(\text{rear} + 1 == \text{front})$
 - ☒ C. $(\text{rear} == \text{front} - 1)$
 - ☒ D. $(\text{rear} == \text{front})$

Question ID : 441009271221

Option 1 ID : 4410091053721

Option 2 ID : 4410091053719

Option 3 ID : 4410091053720

Option 4 ID : 4410091053722

Status : Answered

Chosen Option : A

Q.7 Which of the following is true about IEEE standard floating-point formats?

- Ans**
- ☒ A. Single-precision uses 32 bits, double-precision uses 64 bits
 - ☐ B. Single-precision uses 64 bits, double-precision uses 32 bits
 - ☐ C. Both single and double precision use 32 bits
 - ☐ D. Single-precision uses 16 bits, double-precision uses 32 bits

Question ID : 4410091431235

Option 1 ID : 4410095653231

Option 2 ID : 4410095653230

Option 3 ID : 4410095653232

Option 4 ID : 4410095653233

Status : Answered

Chosen Option : A

Q.8 A graduate student wants to find the likelihood of passing a Statistics exam, given that they attended all lectures. Which formula should the student use to determine the probability of passing given attendance?

- Ans**
- ☐ A. The formula $P(\text{Pass})$ plus $P(\text{Attend})$ is appropriate for this scenario.
 - ☒ B. The formula $P(\text{Pass}|\text{Attend}) = P(\text{Pass and Attend})$ divided by $P(\text{Attend})$ is appropriate.
 - ☐ C. The formula $P(\text{Attend}|\text{Pass}) = P(\text{Pass and Attend})$ divided by $P(\text{Pass})$ is appropriate.
 - ☐ D. The formula $P(\text{Pass})$ times $P(\text{Attend})$ is appropriate for this scenario.

Question ID : 4410092231169

Option 1 ID : 4410098778111

Option 2 ID : 4410098778112

Option 3 ID : 4410098778114

Option 4 ID : 4410098778113

Status : Answered

Chosen Option : B

Q.9 What is the purpose of the Root Name Server in the DNS hierarchy?

- Ans**
- ☒ A. It provides IP addresses of Authoritative Name Servers for Top-Level Domains (TLDs)
 - ☐ B. It resolves domain names directly
 - ☐ C. It caches DNS records for fast access
 - ☐ D. It assigns domain names

Question ID : 44100932857

Option 1 ID : 441009130653

Option 2 ID : 441009130652

Option 3 ID : 441009130654

Option 4 ID : 441009130655

Status : Answered

Chosen Option : A

Q.10 Which of the following best describes an empty set?

Ans ☒ A. A set that has one element

☒ B. A set that contains all elements from a given universal set

☒ C. A set that contains zero as an element

☒ D. A set that has no elements

Question ID : 4410092319316

Option 1 ID : 4410099121696

Option 2 ID : 4410099121695

Option 3 ID : 4410099121693

Option 4 ID : 4410099121694

Status : Answered

Chosen Option : D

Q.11 In the context of algorithm design and analysis, which of the following represents the initial step in the design process?

Ans ☒ A. Analyze the algorithm time complexity

☒ B. Implement the algorithm in a Computer programming language

☒ C. Write the pseudocode

☒ D. Define the problem clearly and specify the input output

Question ID : 4410091237207

Option 1 ID : 4410094882384

Option 2 ID : 4410094882387

Option 3 ID : 4410094882385

Option 4 ID : 4410094882386

Status : Answered

Chosen Option : D

Q.12 Given a code snippet in C programming language as follows:

```
int a = 10;  
int *p;  
int **w;  
p = &a;
```

which statement correctly assigns the address of 'p' to 'w'?

Ans ☒ A. w = &p;

☐ B. w = &&a;

☐ C. w = p;

☐ D. w = *p;

Question ID : 4410092215464

Option 1 ID : 4410098717996

Option 2 ID : 4410098717995

Option 3 ID : 4410098717994

Option 4 ID : 4410098717997

Status : Answered

Chosen Option : A

Q.13 LR Parsing is known for being:

Ans ☐ A. Left-recursive only

☐ B. Non-deterministic always

☐ C. Top-down parsing

☒ D. Bottom-up parsing

Question ID : 4410091381377

Option 1 ID : 4410095456709

Option 2 ID : 4410095456710

Option 3 ID : 4410095456708

Option 4 ID : 4410095456711

Status : Answered

Chosen Option : D

Q.14 Which constraint ensures that a foreign key value in one table either matches a primary key in another table or is NULL (if allowed) in DBMS?

Ans ☒ A. Referential integrity

☐ B. Key constraint

☐ C. Domain constraint

☐ D. Entity integrity

Question ID : 4410091432703

Option 1 ID : 4410095659300

Option 2 ID : 4410095659301

Option 3 ID : 4410095659299

Option 4 ID : 4410095659298

Status : Answered

Chosen Option : A

Q.15 Consider a finite state transducer (FST) with $Q = \{q_0, q_1\}$, $\Sigma = \{0, 1\}$, $\Gamma = \{a, b, c\}$, initial state q_0 , and the following transitions:

$\delta(q_0, 0) = q_1$, $\delta(q_0, 1) = q_0$

$\delta(q_1, 0) = q_0$, $\delta(q_1, 1) = q_1$

and output function:

$\theta(q_0, 0) = a$, $\theta(q_0, 1) = c$

$\theta(q_1, 0) = b$, $\theta(q_1, 1) = a$

What will be the output string for the input 1010?

Ans ☐ A. caba

☐ B. abac

☒ C. caab

☐ D. acab

Question ID : 4410091393259

Option 1 ID : 4410095502675

Option 2 ID : 4410095502674

Option 3 ID : 4410095502677

Option 4 ID : 4410095502676

Status : Answered

Chosen Option : A

Q.16 Which of the following best describes the relationship between pointers and 1D array declaration in C programming language?

Ans ☒ A. The array name points to the first element, but is not a modifiable pointer.

☐ B. The array name is a pointer variable and can be reassigned to a new address

☐ C. The array name and any pointer variable are interchangeable.

☐ D. The array name points to the last element of the array.

Question ID : 4410092215468

Option 1 ID : 4410098718010

Option 2 ID : 4410098718012

Option 3 ID : 4410098718011

Option 4 ID : 4410098718013

Status : Answered

Chosen Option : A

Q.17 A system must recognize any number of consecutive 'a's. Which regular expression correctly models this?

Ans ☐ A. a+

☐ B. a?

☐ C. ab*

☒ D. a*

Question ID : 4410091377043

Option 1 ID : 4410095439720

Option 2 ID : 4410095439722

Option 3 ID : 4410095439721

Option 4 ID : 4410095439719

Status : Answered

Chosen Option : A

Q.18 Which of the following is not a property of a discrete random variable X, that takes any value from 0, 1, 2,...,n?

Ans ☐ A. $\sum P(X = x) = 1$, the summation is taken over all possible values of X

☐ B. For any possible value of X, $P(X = x) \geq 0$

☒ C. $P(X = n) < 0$

☐ D. If A and B are two disjoint subsets of $\{0, 1, 2, \dots, n\}$ then $P(A + B) = P(A) + P(B)$

Question ID : 4410092231174

Option 1 ID : 4410098778133

Option 2 ID : 4410098778131

Option 3 ID : 4410098778132

Option 4 ID : 4410098778134

Status : Answered

Chosen Option : C

Q.19 In computer networks, what is the primary difference between a **service** and a **protocol**?

- Ans**
- ☒ A. A protocol defines what operations are performed; a service defines how they are performed.
 - ☒ B. A service is used by lower layers, while a protocol belongs to higher layers.
 - ☒ C. Both service and protocol define how operations are performed.
 - ☒ D. A service defines what operations are provided; a protocol defines how they are performed.

Question ID : 4410091480341

Option 1 ID : 4410095848451

Option 2 ID : 4410095848453

Option 3 ID : 4410095848452

Option 4 ID : 4410095848450

Status : Answered

Chosen Option : D

Q.20 Which of the following is a valid declaration for a 2D array in C?

- Ans**
- ☒ A. `int a[] = {{1,2}, {3,4}};`
 - ☒ B. `int a[2,2] = {{1,2}, {3,4}};`
 - ☒ C. `array a[2][2] = {{1,2}, {3,4}};`
 - ☒ D. `int a[2][2] = {{1,2}, {3,4}};`

Question ID : 441009234439

Option 1 ID : 441009916178

Option 2 ID : 441009916179

Option 3 ID : 441009916181

Option 4 ID : 441009916180

Status : Answered

Chosen Option : D

Q.21 Consider a Mealy machine designed to process a sequence representing the presence of mineral samples ('M') and non-mineral samples ('N') during a geological survey. The machine outputs 'A' when the current sample is 'M' and the previous sample was 'N', otherwise it outputs 'B'. Which description below correctly reflects the machine's output behavior?

- Ans**
- ☒ A. It produces output based only on the current input symbol.
 - ☒ B. It waits until the next transition to output a value.
 - ☒ C. It changes output only when the input switches from 'M' to 'N'.
 - ☒ D. It produces output based on both the current input and the previous state.

Question ID : 4410092215504

Option 1 ID : 4410098718157

Option 2 ID : 4410098718155

Option 3 ID : 4410098718154

Option 4 ID : 4410098718156

Status : Answered

Chosen Option : C

Q.22 Which of the following is the correct symbol for representing the decimal number 12 in hexadecimal system?

- Ans**
- ☒ A. A
 - ☒ B. C
 - ☒ C. D
 - ☒ D. B

Question ID : 44100978208

Option 1 ID : 441009310840

Option 2 ID : 441009310842

Option 3 ID : 441009310843

Option 4 ID : 441009310841

Status : Answered

Chosen Option : C

Q.23 What we called to the part of an instruction that specifies the operation to be performed?

- Ans
- ☒ A. Operand
 - ☒ B. Label
 - ☒ C. Opcode
 - ☒ D. Comment

Question ID : 4410091427591

Option 1 ID : 4410095638866

Option 2 ID : 4410095638868

Option 3 ID : 4410095638867

Option 4 ID : 4410095638869

Status : Answered

Chosen Option : C

Q.24 Which of the following statements correctly differentiates **connection-oriented** from **connectionless** services?

- Ans
- ☒ A. Connectionless has higher overhead; connection-oriented has minimal.
 - ☒ B. Connection-oriented ensures reliability and order; connectionless may not.
 - ☒ C. Both perform error checking and retransmission.
 - ☒ D. Connection-oriented sends data without a connection; connectionless requires one.

Question ID : 4410091358661

Option 1 ID : 4410095366806

Option 2 ID : 4410095366805

Option 3 ID : 4410095366807

Option 4 ID : 4410095366804

Status : Answered

Chosen Option : A

Q.25 What does each row in a relational table represent?

- Ans
- ☒ A. Tuple
 - ☒ B. Column
 - ☒ C. Domain
 - ☒ D. Attribute

Question ID : 4410091381089

Option 1 ID : 4410095455550

Option 2 ID : 4410095455551

Option 3 ID : 4410095455552

Option 4 ID : 4410095455549

Status : Answered

Chosen Option : D

Q.26 Which of the following traversals can be used to reconstruct a unique binary tree, if combined with Preorder?

- Ans**
- ☒ A. Postorder
 - ☒ B. Level Order
 - ☒ C. Preorder alone is enough
 - ☒ D. Inorder

Question ID : 441009672448

Option 1 ID : 4410092643894

Option 2 ID : 4410092643895

Option 3 ID : 4410092643897

Option 4 ID : 4410092643896

Status : Answered

Chosen Option : A

Q.27 Which statement best describes both an advantage and a disadvantage of Insertion Sort?

- Ans**
- ☒ A. Performs optimally for any input type but lacks stability, failing to preserve the relative order of identical elements.
 - ☒ B. Efficient for nearly sorted data due to adaptability but impractical for large datasets due to $O(n^2)$ complexity.
 - ☒ C. Fastest sorting algorithm for all inputs but requires excessive auxiliary space, making it unsuitable for memory-limited systems.
 - ☒ D. Guarantees logarithmic time for sorted inputs but remains inefficient for small datasets as its complexity stays $O(n^2)$.

Question ID : 44100944938

Option 1 ID : 441009178975

Option 2 ID : 441009178973

Option 3 ID : 441009178974

Option 4 ID : 441009178976

Status : Answered

Chosen Option : B

Q.28 Given a Mealy machine where state q on input 1 outputs 0 and transitions to state p, converting to a Moore machine requires:-

Ans ☒ A. Outputting 0 immediately on entering ppp

☒ B. Splitting p into two states with different outputs

☒ C. No change is needed

☒ D. Adding epsilon transitions

Question ID : 441009876314

Option 1 ID : 4410093458868

Option 2 ID : 4410093458871

Option 3 ID : 4410093458869

Option 4 ID : 4410093458870

Status : Answered

Chosen Option : B

Q.29 Which parsing method uses a stack to store symbols and reduces them when a right-hand side pattern is recognized?

Ans ☒ A. Predictive Parsing

☒ B. Top-Down Parsing

☒ C. Shift-Reduce Parsing

☒ D. Recursive Descent Parsing

Question ID : 4410091377155

Option 1 ID : 4410095440175

Option 2 ID : 4410095440178

Option 3 ID : 4410095440177

Option 4 ID : 4410095440176

Status : Answered

Chosen Option : B

Q.30 What do you mean by TLD name servers?

Ans ☒ A. Top-level domain name servers

☒ B. Tiered-level domain name servers

☒ C. Task-levelled data name servers

☒ D. Top-level disk name servers

Question ID : 4410091439587

Option 1 ID : 4410095687012

Option 2 ID : 4410095687013

Option 3 ID : 4410095687014

Option 4 ID : 4410095687015

Status : Answered

Chosen Option : B

Q.31 In an associative memory system, the argument register is used to:

- Ans**
- ☒ A. Store the result of a match
 - ☒ B. Hold data to be written into memory
 - ☒ C. Generate memory addresses sequentially
 - ☒ D. Compare with key fields in all memory locations simultaneously

Question ID : 441009220121

Option 1 ID : 441009863451

Option 2 ID : 441009863452

Option 3 ID : 441009863454

Option 4 ID : 441009863453

Status : Answered

Chosen Option : B

Q.32 What is the space complexity of an algorithm that uses only a constant amount of memory regardless of input size?

- Ans**
- ☒ A. $O(n)$
 - ☒ B. $O(\log n)$
 - ☒ C. $O(1)$
 - ☒ D. $O(n^2)$

Question ID : 44100977522

Option 1 ID : 441009308022

Option 2 ID : 441009308023

Option 3 ID : 441009308021

Option 4 ID : 441009308024

Status : Answered

Chosen Option : C

Q.33 What is the best approach to delete an element from an unsorted array?

- Ans**
- ☒ A. Replace the element with the last element and reduce the array size
 - ☒ B. Replace the element with 0
 - ☒ C. Sort the array and then delete
 - ☒ D. Shift all elements left after deletion

Question ID : 441009260688

Option 1 ID : 4410091016169

Option 2 ID : 4410091016168

Option 3 ID : 4410091016170

Option 4 ID : 4410091016167

Status : Answered

Chosen Option : C

Q.34 What is the primary function of a Finite State Machine (FSM) simulator?

- Ans**
- ☐ A. To reduce the number of states in FSMs without running them
 - ☐ B. To convert FSMs into executable machine-level programs
 - ☒ C. To visually or textually test and analyze FSM behavior based on inputs and transitions
 - ☐ D. To transform FSMs directly into digital circuit layouts

Question ID : 441009909550

Option 1 ID : 4410093592086

Option 2 ID : 4410093592083

Option 3 ID : 4410093592084

Option 4 ID : 4410093592085

Status : Answered

Chosen Option : C

Q.35 In a shift-reduce parser, which of the following lists includes all four possible parser actions?

- Ans**
- ☐ A. Shift, Jump, Reduce, Halt
 - ☐ B. Reduce, Copy, Accept, Abort
 - ☐ C. Shift, Move, Accept, Reject
 - ☒ D. Shift, Reduce, Accept, Error

Question ID : 4410091398442

Option 1 ID : 4410095523522

Option 2 ID : 4410095523520

Option 3 ID : 4410095523519

Option 4 ID : 4410095523521

Status : Answered

Chosen Option : B

Q.36 Which of the following statements best reflects a comprehensive layered design issue in computer networks?

- Ans**
- ☒ A. Layered design primarily concerns selecting the physical transmission medium and ensuring minimal propagation delay across all layers.
 - ☒ B. Each layer must handle key issues like addressing, error control, flow control, sequencing and data direction, ensuring reliable communication between entities without breaking layer abstraction.
 - ☒ C. Once error control and sequencing are implemented at the data link layer, upper layers are exempt from such design considerations.
 - ☒ D. The design issues of addressing and error control are handled exclusively by the network layer, while flow control and sequencing are managed solely by the transport layer.

Question ID : 4410091445827

Option 1 ID : 4410095711949

Option 2 ID : 4410095711950

Option 3 ID : 4410095711952

Option 4 ID : 4410095711951

Status : Answered

Chosen Option : D

Q.37 Define thrashing in an operating system.

- Ans**
- ☒ A. Excessive swapping of pages in and out of memory
 - ☒ B. A deadlock occurring in memory allocation
 - ☒ C. A read-write operation failure
 - ☒ D. A process being started abnormally

Question ID : 44100974879

Option 1 ID : 441009297797

Option 2 ID : 441009297795

Option 3 ID : 441009297796

Option 4 ID : 441009297794

Status : Answered

Chosen Option : A

Q.38 In a circular queue implemented using an array, how is the rear counter updated after an enqueue operation?

- Ans
- ☒ A. No update to rear pointer is needed in the enqueue operation
 - ☒ B. The rear counter is incremented but never wrapped around to the start
 - ☒ C. Incremented and wrapped around to the start if needed
 - ☒ D. Reset to zero unconditionally

Question ID : 4410092215479

Option 1 ID : 4410098718055

Option 2 ID : 4410098718056

Option 3 ID : 4410098718054

Option 4 ID : 4410098718057

Status : Answered

Chosen Option : A

Q.39 A 32-bit IEEE 754 single-precision floating-point number is represented as:
0100 0001 0100 1001 0000 0000 0000 0000
What is the **decimal value** of this floating-point number?

- Ans
- ☒ A. 25.125
 - ☒ B. 100.5
 - ☒ C. 12.5625
 - ☒ D. 50.25

Question ID : 4410091434964

Option 1 ID : 4410095668394

Option 2 ID : 4410095668396

Option 3 ID : 4410095668393

Option 4 ID : 4410095668395

Status : Answered

Chosen Option : B

Q.40 Which of the following correctly describes an SQL schema?

- Ans
- ☒ A. An SQL schema automatically enforces all foreign key constraints across the database.
 - ☒ B. An SQL schema is identified by a name, includes an authorization identifier for its owner, and contains descriptors for elements.
 - ☒ C. An SQL schema is a table that stores all user data and constraints.
 - ☒ D. An SQL schema is a view that aggregates data from multiple tables.

Question ID : 4410091398549

Option 1 ID : 4410095523942

Option 2 ID : 4410095523940

Option 3 ID : 4410095523939

Option 4 ID : 4410095523941

Status : Answered

Chosen Option : D

Q.41 Which technique is commonly used by a compiler to recognize identifiers in a programming language?

- Ans**
- ☒ A. Manual code scanning
 - ☒ B. Context-free grammar
 - ☒ C. Regular expressions and finite automata
 - ☒ D. High-level code optimization

Question ID : 4410091377116

Option 1 ID : 4410095440021

Option 2 ID : 4410095440020

Option 3 ID : 4410095440019

Option 4 ID : 4410095440022

Status : Answered

Chosen Option : C

Q.42 What will happen if a process with a higher priority arrives in the ready queue in preemptive scheduling algorithm?

- Ans**
- ☒ A. The scheduler compares the total execution times of all processes.
 - ☒ B. The CPU switches to the new process only if it has a smaller burst time.
 - ☒ C. The currently running process is immediately removed from the CPU.
 - ☒ D. It pauses for the current process to finish its CPU burst.

Question ID : 4410091337130

Option 1 ID : 4410095280290

Option 2 ID : 4410095280289

Option 3 ID : 4410095280288

Option 4 ID : 4410095280287

Status : Answered

Chosen Option : C

Q.43 In an associative memory system, the key register holds:

- Ans**
- ☒ A. The memory address of the required data.
 - ☒ B. The tag field used for parallel comparison.
 - ☒ C. The binary result of address decoding.
 - ☒ D. The control bits for memory timing.

Question ID : 4410091385230

Option 1 ID : 4410095471662

Option 2 ID : 4410095471661

Option 3 ID : 4410095471663

Option 4 ID : 4410095471660

Status : Answered

Chosen Option : A

Q.44 If a system uses 32-bit virtual addresses and a page size of 4 KB, how many entries are needed in a single-level page table?

Ans ☒ A. 2^{12}

☒ B. 2^{20}

☒ C. 2^{10}

☒ D. 2^{32}

Question ID : 4410091433455

Option 1 ID : 4410095662385

Option 2 ID : 4410095662383

Option 3 ID : 4410095662382

Option 4 ID : 4410095662384

Status : Answered

Chosen Option : B

Q.45 A team develops a text-mining tool that highlights sequences in security logs where a server's alert code is immediately followed by an IP address in the format 10.0.X.Y (where X and Y are integers). Which regular expression most efficiently identifies these patterns without false positives?

Ans ☒ A. `alert\s10\.0\.[0-9]{1,3}\.[A-Z]+`

☒ B. `alert10\.0\.[0-9]+\.[0-9]+`

☒ C. `alert\s10\.0\.[0-9]*\.[0-9]*`

☒ D. `alert\s10\.0\.[0-9]+\.[0-9]+`

Question ID : 4410092215509

Option 1 ID : 4410098718176

Option 2 ID : 4410098718174

Option 3 ID : 4410098718175

Option 4 ID : 4410098718177

Status : Answered

Chosen Option : D

Q.46 Which of the following best describes the memory hierarchy in a computer system?

Ans ☒ A. Main Memory → Registers → Cache → Secondary Storage

☒ B. Registers → Cache → Main Memory → Secondary Storage

☒ C. Cache → Main Memory → Registers → Secondary Storage

☒ D. Secondary Storage → Main Memory → Cache → Registers

Question ID : 441009752155

Option 1 ID : 4410092962537

Option 2 ID : 4410092962535

Option 3 ID : 4410092962534

Option 4 ID : 4410092962536

Status : Answered

Chosen Option : D

Q.47 What is the initial value of top in an empty stack implemented using an array?

- Ans**
- ☒ A. 0
- ☒ B. Depends on stack size
- ☒ C. 1
- ☒ D. -1

Question ID : 44100976729

Option 1 ID : 441009305057

Option 2 ID : 441009305060

Option 3 ID : 441009305058

Option 4 ID : 441009305059

Status : Answered

Chosen Option : D

Q.48 During a content search operation in associative memory, which register temporarily holds the key used to compare against all stored memory words simultaneously?

- Ans**
- ☒ A. Accumulator
- ☒ B. Data Register
- ☒ C. Argument Register
- ☒ D. Program Counter

Question ID : 4410091375130

Option 1 ID : 4410095431829

Option 2 ID : 4410095431826

Option 3 ID : 4410095431828

Option 4 ID : 4410095431827

Status : Answered

Chosen Option : D

Q.49 What is the division of the main memory in Fixed Partitioning in memory management of operating system?

- Ans**
- ☒ A. Dynamically resized when a process terminates.
- ☒ B. Made at boot time and remain unchanged in size and number.
- ☒ C. Destroyed as needed by the processes.
- ☒ D. Created as needed by the processes.

Question ID : 4410091337155

Option 1 ID : 4410095280410

Option 2 ID : 4410095280409

Option 3 ID : 4410095280408

Option 4 ID : 4410095280407

Status : Answered

Chosen Option : D

Q.50 In paging, the page table is used to:

- Ans** ☒ A. Map logical pages to physical frames
- ☐ B. Handle external fragmentation
- ☐ C. Store the physical addresses of all processes
- ☐ D. Keep track of free frames only

Question ID : 4410091433461

Option 1 ID : 4410095662404

Option 2 ID : 4410095662406

Option 3 ID : 4410095662403

Option 4 ID : 4410095662405

Status : Answered

Chosen Option : A

Q.51 Consider a binary search tree (BST). Which one of the following properties is always true for an inorder traversal of the BST?

- Ans** ☒ A. It gives the nodes in ascending order of values of keys
- ☐ B. It gives the nodes in random order of values of keys
- ☐ C. It only visits the leftmost and rightmost nodes
- ☐ D. It gives the nodes in descending order of values of keys

Question ID : 441009750032

Option 1 ID : 4410092954004

Option 2 ID : 4410092954003

Option 3 ID : 4410092954005

Option 4 ID : 4410092954002

Status : Answered

Chosen Option : A

Q.52 The minimized Deterministic Finite State Machine (DFSM) in theory of computation for a language always has:

- Ans** ☐ A. Equal number of final and non-final states
- ☒ B. The minimum number of states possible for that language
- ☐ C. Exactly one final state
- ☐ D. The same number of states as the NFA

Question ID : 4410091210296

Option 1 ID : 4410094774688

Option 2 ID : 4410094774687

Option 3 ID : 4410094774686

Option 4 ID : 4410094774689

Status : Answered

Chosen Option : B

Q.53 Which of the following correctly represents the cofactor expansion of a 3x3 determinant along the first row?

- Ans** ☒ A. Sum of each element of the first row times its corresponding cofactor
- ☒ B. Sum of only diagonal elements times their cofactors
- ☒ C. Sum of elements of the first column times their cofactors
- ☒ D. Difference of elements of the first row and their cofactors

Question ID : 4410092319372

Option 1 ID : 4410099121918

Option 2 ID : 4410099121920

Option 3 ID : 4410099121917

Option 4 ID : 4410099121919

Status : Answered

Chosen Option : C

Q.54 If a measurement is recorded as 12.8 cm and the true value is 13.0 cm, what is the absolute error?

- Ans** ☒ A. 1.2 cm
- ☒ B. 0 cm
- ☒ C. 0.2 cm
- ☒ D. 0.1 cm

Question ID : 4410092319380

Option 1 ID : 4410099121949

Option 2 ID : 4410099121950

Option 3 ID : 4410099121952

Option 4 ID : 4410099121951

Status : Answered

Chosen Option : B

Q.55 Which of the following is true about Selection Sort?

- Ans** ☒ A. It is an adaptive sorting algorithm.
- ☒ B. It has $O(n^2)$ time complexity in all cases.
- ☒ C. It performs well on large datasets.
- ☒ D. It is a stable sorting algorithm.

Question ID : 44100978136

Option 1 ID : 441009310533

Option 2 ID : 441009310532

Option 3 ID : 441009310531

Option 4 ID : 441009310530

Status : Answered

Chosen Option : B

Q.56 What shape is commonly used to represent an entity in an ER diagram?

- Ans**
- ☐ A. Triangle
 - ☐ B. Diamond
 - ☐ C. Ellipse
 - ☒ D. Rectangle

Question ID : 4410091381085

Option 1 ID : 4410095455533

Option 2 ID : 4410095455536

Option 3 ID : 4410095455534

Option 4 ID : 4410095455535

Status : Answered

Chosen Option : C

Q.57 Which of the following statements best distinguishes the relationship between a service and a protocol in a layered network architecture?

- Ans**
- ☐ A. A service is the physical implementation of a protocol; thus, changing the protocol automatically changes the provided service.
 - ☒ B. A service specifies what operations a layer offers to the layer above, whereas a protocol defines how peer entities within the same layer communicate to provide that service.
 - ☐ C. Both services and protocols define the message formats and meanings used between the same layer entities, but at different abstraction levels.
 - ☐ D. A protocol defines the interface between two adjacent layers, while a service specifies the message exchange rules between peer entities.

Question ID : 4410091445831

Option 1 ID : 4410095711964

Option 2 ID : 4410095711962

Option 3 ID : 4410095711963

Option 4 ID : 4410095711961

Status : Answered

Chosen Option : B

Q.58 Consider the following grammar **G**:

$E \rightarrow T E'$
 $E' \rightarrow + T E' \epsilon$
 $T \rightarrow F T'$
 $T' \rightarrow * F T' \epsilon$
 $F \rightarrow (E) \text{id}$

Which of the following statements about **top-down parsing** of grammar **G** is true?

- Ans**
- ☒ A. The grammar is suitable for LL(1) (predictive) parsing and does not require backtracking.
 - ☐ B. The grammar is not left-recursive and can be parsed using a predictive (LL(1)) parser.
 - ☐ C. The grammar requires backtracking since FIRST and FOLLOW sets of E' overlap.
 - ☐ D. The grammar is left-recursive and hence cannot be used for top-down parsing.

Question ID : 4410091440499

Option 1 ID : 4410095690674

Option 2 ID : 4410095690672

Option 3 ID : 4410095690673

Option 4 ID : 4410095690671

Status : Answered

Chosen Option : C

Q.59 What does a node in a singly linked list typically contain?

- Ans**
- ☐ A. A pointer only
 - ☐ B. Data only
 - ☒ C. Data and two pointers
 - ☐ D. Data and one pointer

Question ID : 441009260694

Option 1 ID : 4410091016192

Option 2 ID : 4410091016191

Option 3 ID : 4410091016193

Option 4 ID : 4410091016194

Status : Answered

Chosen Option : D

Q.60 Which of the following is an application of compiler technology?

- Ans
- ☒ A. Cache mapping
 - ☒ B. Disk scheduling
 - ☒ C. Code optimization
 - ☒ D. Virtual memory management

Question ID : 4410091188167

Option 1 ID : 4410094686925

Option 2 ID : 4410094686924

Option 3 ID : 4410094686922

Option 4 ID : 4410094686923

Status : Answered

Chosen Option : A

Q.61 Which statement best describes a zero-address instruction in stack-based architectures?

- Ans
- ☒ A. Operands are implicitly taken from the top of the stack.
 - ☒ B. The instruction contains two explicit operand fields.
 - ☒ C. Operands are stored in accumulator registers.
 - ☒ D. Operands are fetched directly from main memory.

Question ID : 4410091385165

Option 1 ID : 4410095471406

Option 2 ID : 4410095471404

Option 3 ID : 4410095471407

Option 4 ID : 4410095471405

Status : Answered

Chosen Option : D

Q.62 _____ is **NOT** a valid token typically generated by a lexical analyzer.

- Ans
- ☒ A. Parser
 - ☒ B. Variable
 - ☒ C. Compiler
 - ☒ D. Keyword

Question ID : 4410091427603

Option 1 ID : 4410095638916

Option 2 ID : 4410095638915

Option 3 ID : 4410095638914

Option 4 ID : 4410095638917

Status : Answered

Chosen Option : C

Q.63 Which of the following **Windows API functions** is used to write data to a file?

- Ans ☒ A. fwrite()
- ☒ B. WriteFile()
- ☒ C. CreateFile()
- ☒ D. fprintf()

Question ID : 4410091186134

Option 1 ID : 4410094678833

Option 2 ID : 4410094678832

Option 3 ID : 4410094678831

Option 4 ID : 4410094678834

Status : Answered

Chosen Option : A

Q.64 Given two random variables X and Y with $E(X) = 2$, $E(Y) = 3$, and $E(XY) = 10$, what is the covariance of X and Y?

- Ans ☒ A. 6
- ☒ B. 5
- ☒ C. 1
- ☒ D. 4

Question ID : 4410092319377

Option 1 ID : 4410099121938

Option 2 ID : 4410099121940

Option 3 ID : 4410099121937

Option 4 ID : 4410099121939

Status : Answered

Chosen Option : C

Q.65 Which of the following statements is true regarding the differentiability of a function at a point?

- Ans ☒ A. A function must be continuous at that point to be differentiable there.
- ☒ B. A function is differentiable at a point only if its derivative is infinite there.
- ☒ C. A function may be discontinuous at a point and still be differentiable there.
- ☒ D. A function must be differentiable everywhere to be continuous at a point.

Question ID : 4410092319369

Option 1 ID : 4410099121906

Option 2 ID : 4410099121905

Option 3 ID : 4410099121907

Option 4 ID : 4410099121908

Status : Answered

Chosen Option : A

Q.66 You are tasked to construct a binary search tree (BST) from a stream of incoming sensor readings, where each reading is processed as it arrives. Midway, the sensor malfunctions and emits a duplicate reading not originally in the stream. To preserve BST properties and ensure efficient future search operations, which strategy should you employ for the duplicate reading?

- Ans**
- ☒ A. Replace the existing node with the duplicate value and continue
- ☒ B. Allow the duplicate in the left subtree and continue
- ☒ C. Allow the duplicate in the right subtree and continue
- ☒ D. Ignore the duplicate and continue inserting unique values

Question ID : 4410092215489

Option 1 ID : 4410098718097

Option 2 ID : 4410098718094

Option 3 ID : 4410098718095

Option 4 ID : 4410098718096

Status : Answered

Chosen Option : B

Q.67 Which operation in Relational Algebra is used to retrieve specific columns from a relation?

- Ans**
- ☒ A. Intersection ($\cap$)
- ☒ B. Cartesian Product ($\times$)
- ☒ C. Selection (σ)
- ☒ D. Projection (π)

Question ID : 44100930927

Option 1 ID : 441009123199

Option 2 ID : 441009123198

Option 3 ID : 441009123196

Option 4 ID : 441009123197

Status : Answered

Chosen Option : D

Q.68 The entity integrity constraint ensures that:

- Ans**
- ☒ A. All attribute values must belong to the same domain
- ☒ B. Each tuple must reference at least one other tuple
- ☒ C. No primary key value can be NULL
- ☒ D. No foreign key value can be NULL

Question ID : 4410091316882

Option 1 ID : 4410095199088

Option 2 ID : 4410095199087

Option 3 ID : 4410095199089

Option 4 ID : 4410095199086

Status : Answered

Chosen Option : A

Q.69 A logistics company manages a tracking system where each package ID is stored in a chronological list as the packages are processed. An engineer suggests that, to locate a particular package's ID, the system should check each ID sequentially from the start until the target is found or the list ends. Which statement most accurately reflects the main drawback of this approach if the company expects the number of packages to increase dramatically over time?

- Ans**
- ☐ A. The time to search for any package ID will stay about the same, regardless of list size.
 - ☐ B. The memory needed to store package IDs will remain constant even as the list grows.
 - ☒ C. Search time grows linearly with the number of packages.
 - ☐ D. Finding a package ID will become faster as the number of packages increases.

Question ID : 4410092215498

Option 1 ID : 4410098718133

Option 2 ID : 4410098718132

Option 3 ID : 4410098718130

Option 4 ID : 4410098718131

Status : Answered

Chosen Option : C

Q.70 Which of the following is a necessary condition for a function $f: A \rightarrow B$ to be invertible?

- Ans**
- ☐ A. f must be only onto
 - ☒ B. f must be both one-to-one and onto
 - ☐ C. f must be only one-to-one
 - ☐ D. f can be neither one-to-one nor onto

Question ID : 4410092319300

Option 1 ID : 4410099121632

Option 2 ID : 4410099121630

Option 3 ID : 4410099121631

Option 4 ID : 4410099121629

Status : Answered

Chosen Option : B

Q.71 What is the worst-case time complexity of a linear search algorithm in an unsorted array of n elements?

- Ans**
- ☐ A. $O(n \log n)$
- ☐ B. $O(1)$
- ☒ C. $O(n)$
- ☐ D. $O(\log n)$

Question ID : 441009749075

Option 1 ID : 4410092950125

Option 2 ID : 4410092950127

Option 3 ID : 4410092950126

Option 4 ID : 4410092950124

Status : Answered

Chosen Option : C

Q.72 Which of the following statements are TRUE about server and client side scripting in computer networks?

S1: PHP is used for server side scripting
 S2: HTML is used for database operations
 S3: CSS is used for server side scripting
 S4: ASP.NET is used for server side scripting

- Ans**
- ☐ A. Only S2 and S3
- ☐ B. Only S1, S2 and S4
- ☒ C. Only S1 and S4
- ☐ D. Only S1 and S3

Question ID : 441009324260

Option 1 ID : 4410091263270

Option 2 ID : 4410091263268

Option 3 ID : 4410091263269

Option 4 ID : 4410091263267

Status : Answered

Chosen Option : C

Q.73 Which of the following options about non-preemptive scheduling is incorrect?

- Ans**
- ☒ A. Non-preemptive scheduling can lead to process starvation.
 - ☒ B. Round Robin is classified as a non-preemptive scheduling algorithm.
 - ☒ C. In non-preemptive scheduling, the CPU is relinquished only when a process terminates or initiates an input/output operation.
 - ☒ D. In non-preemptive scheduling, the response time and waiting time are generally not equal for all processes, and in most cases, waiting time is greater than response time.

Question ID : 4410091423907

Option 1 ID : 4410095624438

Option 2 ID : 4410095624436

Option 3 ID : 4410095624437

Option 4 ID : 4410095624439

Status : Answered

Chosen Option : B

Q.74 Which statement correctly differentiates between SRAM and DRAM?

- Ans**
- ☒ A. Both have the same speed and cost
 - ☒ B. DRAM is faster and more expensive than SRAM
 - ☒ C. SRAM is faster but more expensive than DRAM
 - ☒ D. SRAM requires periodic refreshing; DRAM does not

Question ID : 4410091396074

Option 1 ID : 4410095513955

Option 2 ID : 4410095513953

Option 3 ID : 4410095513954

Option 4 ID : 4410095513952

Status : Answered

Chosen Option : B

Q.75 What is the output of a JK flip-flop when both inputs J = 1 and K = 1?

- Ans**
- ☒ A. Output toggles
 - ☒ B. Output remains unchanged
 - ☒ C. Output is set to 1
 - ☒ D. Output is set to 0

Question ID : 441009872029

Option 1 ID : 4410093441293

Option 2 ID : 4410093441292

Option 3 ID : 4410093441291

Option 4 ID : 4410093441290

Status : Answered

Chosen Option : D

Q.76 If a D Flip-Flop has $D = 1$ and a rising edge of the clock occurs, what will the Q output be with reference to Sequential Circuits?

- Ans**
- ☒ A. 0
 - ☒ B. Undefined
 - ☒ C. Toggle
 - ☒ D. 1

Question ID : 4410091431266

Option 1 ID : 4410095653354

Option 2 ID : 4410095653357

Option 3 ID : 4410095653356

Option 4 ID : 4410095653355

Status : Answered

Chosen Option : D

Q.77 Which of the following is not a connection-oriented service?

- Ans**
- ☒ A. Perform a Handshake to set up the connection.
 - ☒ B. Stateless packet forwarding.
 - ☒ C. Reliable byte-stream delivery.
 - ☒ D. Maintain frame sequencing.

Question ID : 4410091438628

Option 1 ID : 4410095683215

Option 2 ID : 4410095683213

Option 3 ID : 4410095683214

Option 4 ID : 4410095683216

Status : Answered

Chosen Option : B

Q.78 A list of $n-1$ integers is provided, where each integer is in the range of 1 to n , and there are no duplicates. One integer is missing from the list. What is the time and space complexity if we apply a brute force algorithm to find the missing integer?

- Ans**
- ☒ A. $O(1)$ time, $O(1)$ space
 - ☒ B. $O(n^2)$ time, $O(1)$ space
 - ☒ C. $O(n)$ time, $O(n)$ space
 - ☒ D. $O(n \log n)$ time, $O(1)$ space

Question ID : 44100944867

Option 1 ID : 441009178693

Option 2 ID : 441009178692

Option 3 ID : 441009178694

Option 4 ID : 441009178695

Status : Answered

Chosen Option : B

Q.79 Given the 2D array `int arr[3][3] = {{1,2,3}, {4,5,6}, {7,8,9}};`, what is the value of `*(*(arr + 1) + 2)`?

- Ans
- ☐ A. 3
 - ☐ B. 5
 - ☒ C. 6
 - ☐ D. 8

Question ID : 441009235187

Option 1 ID : 441009919170

Option 2 ID : 441009919172

Option 3 ID : 441009919171

Option 4 ID : 441009919173

Status : Answered

Chosen Option : C

Q.80 Which of the following statements correctly describes the power set of the set {a, b}?

- Ans
- ☐ A. It is the set containing only the original set {a, b}
 - ☒ B. It is the set containing all possible subsets of {a, b}
 - ☐ C. It is the set containing only the singleton subsets of {a, b}
 - ☐ D. It is the set containing only the empty set of {a, b}

Question ID : 4410092319315

Option 1 ID : 4410099121689

Option 2 ID : 4410099121692

Option 3 ID : 4410099121691

Option 4 ID : 4410099121690

Status : Answered

Chosen Option : C

Q.81 Which notation represents the worst-case time complexity?

- Ans
- ☐ A. Small o (o)
 - ☒ B. Big O (O)
 - ☐ C. Big Theta (Θ)
 - ☐ D. Big Omega (Ω)

Question ID : 4410091470075

Option 1 ID : 4410095808229

Option 2 ID : 4410095808228

Option 3 ID : 4410095808227

Option 4 ID : 4410095808226

Status : Answered

Chosen Option : B

Q.82 Which of the following statements correctly describes Isolation and Transaction Isolation Levels in a database system?

Statement-1: Ensuring the isolation property is the responsibility of a component of the database system called the concurrency-control system.
Statement-2: The isolation levels may be Serializable, Repeatable read, Read committed, Read uncommitted and all the isolation levels additionally disallow dirty writes, that is, they disallow writes to a data item that has already been written by another transaction that has not yet committed or aborted.

Ans ☒ A. Statement-1 is True and Statement-2 is True

☐ B. Statement-1 is True and Statement-2 is False

☐ C. Statement-1 is False and Statement-2 is False

☐ D. Statement-1 is False and Statement-2 is True

Question ID : 4410091357939

Option 1 ID : 4410095363920

Option 2 ID : 4410095363917

Option 3 ID : 4410095363919

Option 4 ID : 4410095363918

Status : Answered

Chosen Option : B

Q.83 Which statement about left recursion in context-free grammars is accurate?

Ans ☐ A. Left recursion has no impact on recursive-descent parsing.

☐ B. Left recursion occurs when a non-terminal appears at the end of its own production.

☐ C. Left recursion is required for building LL(1) parsers without any modification.

☒ D. Left recursion can be direct or indirect and may lead to infinite recursion in top-down parsers.

Question ID : 4410091433445

Option 1 ID : 4410095662345

Option 2 ID : 4410095662342

Option 3 ID : 4410095662344

Option 4 ID : 4410095662343

Status : Answered

Chosen Option : C

Q.84 In SRTF (Shortest Remaining Time First) scheduling, which process is selected next?

- Ans** ☒ A. One with the shortest remaining burst time
- ☐ B. One with the longest burst time
- ☐ C. One that arrives first
- ☐ D. One with the highest priority

Question ID : 4410091186159

Option 1 ID : 4410094678889

Option 2 ID : 4410094678887

Option 3 ID : 4410094678888

Option 4 ID : 4410094678890

Status : Answered

Chosen Option : A

Q.85 Which one of the following statements is true for a Hermitian matrix?

- Ans** ☐ A. All elements are positive numbers
- ☐ B. All entries are real numbers
- ☐ C. Sum of diagonal elements is zero
- ☒ D. All diagonal elements are real numbers

Question ID : 4410092319371

Option 1 ID : 4410099121915

Option 2 ID : 4410099121913

Option 3 ID : 4410099121916

Option 4 ID : 4410099121914

Status : Answered

Chosen Option : B

Q.86 Which of the following scenarios best describes a violation of the consistency property?

- Ans** ☐ A. A transaction partially runs due to a crash.
- ☐ B. Two transactions read and write the same record simultaneously.
- ☒ C. A transaction completes but leaves the database in a state where total credits $\neq$ total debits.
- ☐ D. Data from a committed transaction is deleted after a restart.

Question ID : 4410091196225

Option 1 ID : 4410094718596

Option 2 ID : 4410094718597

Option 3 ID : 4410094718599

Option 4 ID : 4410094718598

Status : Answered

Chosen Option : C

Q.87 In the context of programming languages, if a compiler can determine the scope of a declaration by examining the program text alone, the language is said to use:

- Ans**
- ☒ A. Static policy and dynamic scope
 - ☒ B. Dynamic policy and lexical scope
 - ☒ C. Static policy and lexical scope
 - ☒ D. Dynamic policy and dynamic scope

Question ID : 4410091398403

Option 1 ID : 4410095523371

Option 2 ID : 4410095523368

Option 3 ID : 4410095523370

Option 4 ID : 4410095523369

Status : Answered

Chosen Option : A

Q.88 Consider the following declaration of array in C language.

```
int arr[5];
```

Which of the following statements is TRUE about array 'arr'?

- Ans**
- ☒ A. The 'arr' array can store only integer values
 - ☒ B. The 'arr' array can store integers and float values
 - ☒ C. The size of array 'arr' can be increased or decreased during run time dynamically
 - ☒ D. The declaration of the array 'arr' is incorrect and will result in an error

Question ID : 441009249806

Option 1 ID : 441009973157

Option 2 ID : 441009973158

Option 3 ID : 441009973156

Option 4 ID : 441009973159

Status : Answered

Chosen Option : A

Q.89 Why is it considered good practice to close files explicitly even if the programming language runtime may close them at program termination?

- Ans**
- ☒ A. To guarantee file permissions are reset
 - ☒ B. To ensure the file remains hidden and cannot be reopened by others
 - ☒ C. To release resources promptly and avoid resource leaks during execution
 - ☒ D. To protect the file from accidental deletion

Question ID : 4410092215466

Option 1 ID : 4410098718004

Option 2 ID : 4410098718002

Option 3 ID : 4410098718003

Option 4 ID : 4410098718005

Status : Answered

Chosen Option : A

Q.90 A botanist creates a digital representation of a rare plant's evolutionary tree, where each node records a unique genetic mutation. During research, they need to extract the chronological sequence of mutations as they would naturally occur along the main lineage. Which traversal algorithm, when applied to the tree, most accurately reflects the order of these mutations if left sub-branches represent earlier side mutations and right sub-branches represent later side mutations?

- Ans**
- ☒ A. Applying a postorder traversal lists mutations from leaves to root, not main lineage order.
 - ☒ B. A preorder traversal starts with the main mutation, missing proper side branch sequencing.
 - ☒ C. Level order traversal groups mutations by depth, not by evolutionary chronology.
 - ☒ D. Performing an inorder traversal yields the desired mutation sequence along the main lineage.

Question ID : 4410092215495

Option 1 ID : 4410098718118

Option 2 ID : 4410098718120

Option 3 ID : 4410098718119

Option 4 ID : 4410098718121

Status : Answered

Chosen Option : D

Q.91 What makes a finite automaton deterministic?

- Ans**
- ☒ A. It can accept infinite input
 - ☒ B. For each state and input, there is exactly one transition
 - ☒ C. It has no start state
 - ☒ D. It can have multiple transitions for the same input

Question ID : 4410091400927

Option 1 ID : 4410095533317

Option 2 ID : 4410095533316

Option 3 ID : 4410095533314

Option 4 ID : 4410095533315

Status : Answered

Chosen Option : B

Q.92 In the grammar $S \rightarrow aS \mid bS \mid \epsilon$, which strings belong to the language?

- Ans**
- ☒ A. Only "a" and "b"
 - ☒ B. string of "ab" or "ba"
 - ☒ C. Any string of a's and b's
 - ☒ D. Only "ab" and "ba"

Question ID : 4410091479901

Option 1 ID : 4410095846702

Option 2 ID : 4410095846705

Option 3 ID : 4410095846703

Option 4 ID : 4410095846704

Status : Answered

Chosen Option : C

Q.93 Which of the following best describes the interaction between CPU and DMA during block transfer?

- Ans**
- ☒ A. CPU and DMA alternate each word transfer over the bus
 - ☒ B. CPU supervises every byte of transfer to ensure correctness
 - ☒ C. CPU transfers data to cache, DMA transfers it to I/O
 - ☒ D. CPU sends control information once, then DMA handles the transfer independently

Question ID : 4410091345122

Option 1 ID : 4410095313166

Option 2 ID : 4410095313164

Option 3 ID : 4410095313167

Option 4 ID : 4410095313165

Status : Answered

Chosen Option : C

Q.94 Which scenario is least suitable for sparse matrix representation in computer memory?

- Ans**
- ☒ A. A diagonal matrix
 - ☒ B. A large matrix with many zeros
 - ☒ C. An adjacency matrix for a large network
 - ☒ D. A matrix where most elements are nonzero

Question ID : 4410092215470

Option 1 ID : 4410098718018

Option 2 ID : 4410098718019

Option 3 ID : 4410098718021

Option 4 ID : 4410098718020

Status : Answered

Chosen Option : A

Q.95 Which of the following is NOT a common design issue that needs to be addressed in each layer of a computer network?

- Ans**
- ☒ A. Addressing and identifying devices on the network
 - ☒ B. Determining the color scheme of the user interface
 - ☒ C. Data transmission and error control
 - ☒ D. Flow control and sequencing of data

Question ID : 4410091433568

Option 1 ID : 4410095662821

Option 2 ID : 4410095662823

Option 3 ID : 4410095662822

Option 4 ID : 4410095662824

Status : Answered

Chosen Option : B

Q.96 A banking transaction commits successfully, but a system crash occurs immediately afterward. Which mechanism ensures durability?

- Ans**
- ☒ A. Redo log
 - ☒ B. Locking
 - ☒ C. Transaction isolation
 - ☒ D. Partial commit

Question ID : 4410091360411

Option 1 ID : 4410095373706

Option 2 ID : 4410095373707

Option 3 ID : 4410095373705

Option 4 ID : 4410095373708

Status : Answered

Chosen Option : C

Q.97 In a machine instruction, what does the opcode primarily define?

Ans ☒ A. The size of the instruction in bits.

☒ B. The type of operation the processor should execute.

☒ C. The number of registers used in execution.

☒ D. The memory address where data is stored.

Question ID : 44100927964

Option 1 ID : 441009111402

Option 2 ID : 441009111401

Option 3 ID : 441009111403

Option 4 ID : 441009111400

Status : Answered

Chosen Option : B

Q.98 Which statement correctly defines the regular expressions?

Statement-1: Regular expressions representation doesn't involve a combination of strings of symbols from some alphabet Σ , parentheses and the operators $+$, $\times$, and $*$.

Statement-2: A regular expression is not obtained from the symbol $\{a, b, c\}$, empty string ϵ , and empty-set Φ perform the operations $+$, $\times$ and $*$ (union, concatenation and Kleene star).

Ans ☒ A. Statement-1 is False and Statement-2 is True

☒ B. Statement-1 is True and Statement-2 is False

☒ C. Statement-1 is False and Statement-2 is False

☒ D. Statement-1 is True and Statement-2 is True

Question ID : 4410091449312

Option 1 ID : 4410095725695

Option 2 ID : 4410095725694

Option 3 ID : 4410095725696

Option 4 ID : 4410095725697

Status : Answered

Chosen Option : A

Q.99	The hardware organization of associative memory consists of:
Ans	☒ A. Memory Array Logic, Argument Register(A), Key Register(K) and Match Register(M) ☐ B. Argument Register(A), Memory Array Logic and Match Register(M) Only ☐ C. Argument Register(A), Key Register(K) and Match Register(M) Only ☐ D. Memory Array Logic, Key Register(K) and Match Register(M) Only

Question ID : 4410091329507

Option 1 ID : 4410095249580

Option 2 ID : 4410095249579

Option 3 ID : 4410095249577

Option 4 ID : 4410095249578

Status : Answered

Chosen Option : B

Q.100 In a memory system with **fixed-sized partitions**, each holding one process (as in IBM OS/360 MFT). Which statement is **correct**?

Ans	☐ A. Partition size adjusts dynamically ☒ B. Multiprogramming limited; internal fragmentation possible ☐ C. Multiprogramming is unlimited ☐ D. Each partition holds multiple processes
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Question ID : 4410091358411

Option 1 ID : 4410095365799

Option 2 ID : 4410095365798

Option 3 ID : 4410095365796

Option 4 ID : 4410095365797

Status : Answered

Chosen Option : B